Inspection of Fired Boilers and Heaters

API RECOMMENDED PRACTICE 573
FIRST EDITION, OCTOBER 1991

American Petroleum Institute
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Refining Department

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FOREWORD

This recommended practice is based on the accumulated knowledge and experience of engineers and other personnel in the petroleum industry. The information contained in this publication was previously presented as Chapters VIII and IX of the *Guide for Inspection of Refinery Equipment,* which is currently being reorganized as individual recommended practices. The information in this recommended practice does not constitute and should not be construed as a code of rules, regulations, or minimum safe practices. The practices described in this publication are not intended to supplant other practices that have proven satisfactory, nor is this publication intended to discourage innovation and originality in the inspection of refineries. Users of this recommended practice are reminded that no book or manual is a substitute for the judgment of a responsible, qualified person.

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Certain serious adverse health effects are associated with asbestos, among them the serious and often fatal diseases of lung cancer, asbestosis, and mesothelioma (a cancer of the chest and abdominal linings). The degree of exposure to asbestos varies with the product and the work practices involved.

Consult the most recent edition of the Occupational Safety and Health Administration (OSHA), U.S. Department of Labor, Occupational Safety and Health Standard for Asbestos, Tremolite, Anthophyllite, and Actinolite, 29 *Code of Federal Regulations* Section 1910.1001; the U.S. Environmental Protection Agency, National Emission Standard for Asbestos, 40 *Code of Federal Regulations* Sections 61.140 through 61.156; and the U.S. Environmental Protection Agency (EPA) rule on labeling requirements and phased banning of asbestos products, published at 54 *Federal Register* 29460 (July 12, 1989).

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Inspection of Fired Boilers and Heaters

SECTION 1—GENERAL

1.1 Scope

This publication combines the *Guide for Inspection of Refinery Equipment*, Chapter VIII, "Direct-Fired Boilers and Auxiliary Equipment," with Chapter IX, "Fired Heaters and Stacks," and provides guidance on the inspection of fired boilers and heaters. This guidance is meant to promote proactive inspection procedures and to thereby prevent equipment failures and increase overall equipment reliability and plant safety.

1.2 Referenced Publications

To the extent specified in this recommended practice, the most recent edition or revision of the following standards, codes, and specifications shall form a part of this recommended practice:

AISC[1]

M015L *Manual of Steel Construction, Load and Resistance Factor Design*

M016 *Manual of Steel Construction, Allowable Stress Design*

API

RP 530 *Calculation of Heater Tube Thickness in Petroleum Refineries*

Std 560 *Fired Heaters for General Refinery Services*

RP 571 *Conditions Causing Deterioration or Failure* (in press)

RP 572 *Inspection of Pressure Vessels, Heat Exchangers, Condensers, and Coolers* (in press)

ASME[2]

B16.9 *Factory-Made Wrought Steel Buttwelding Fittings*

B16.28 *Wrought Steel Buttwelding Short Radius Elbows and Returns*

B31.1 *Power Piping*

B31.2 *Fuel Gas Piping*

B31.3 *Chemical Plant and Petroleum Refinery Piping*

B31.4 *Liquid Transportation Systems for Hydrocarbons, Liquid Petroleum Gas, Anhydrous Ammonia, and Alcohols*

B31.5 *Refrigeration Piping*

B31.8 *Gas Transmission and Distribution Piping Systems*

B31.9 *Building Services Piping*

B31.11 *Slurry Transportation Piping Systems*

B31 Guide *Corrosion Control for ANSI[3] B31.1 Power Piping Systems*

B31G *Manual for Determining the Remaining Strength of Corroded Pipelines: A Supplement to B31, Code for Pressure Piping*

Boiler and Pressure Vessel Code, Section I, "Power Boilers," Section IV, "Heating Boilers," Section VI, "Recommended Rules for Care and Operation of Heating Boilers," and Section VII, "Recommended Guidelines for Care of Power Boilers"

ASTM[4]

A 297 *Steel Castings, Iron-Chromium and Iron-Chromium-Nickel, Heat-Resistant, for General Application*

NACE[5]

RP01 *Protection of Austenitic Stainless Steel From Polythionic Acid Stress Corrosion Cracking During Shutdown of Refinery Equipment*

1.3 Description

1.3.1 TYPES OF HEATERS

1.3.1.1 General

There are a variety of designs for fired tubular heaters. Some of the more commonly used designs are the box, cylindrical, and cabin designs. Typical heater designs are represented in Figure 1.

1.3.1.2 Box–Type Heaters

A box-type heater is a heater whose structural configuration forms a box. There are many different designs for box-type heaters. These designs involve a variety of tube coil configurations, including horizontal, vertical, helical, and arbor configurations.

[1]American Institute of Steel Construction, 400 North Michigan Avenue, Chicago, Illinois 60611.
[2]American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.
[3]American National Standards Institute, 1430 Broadway, New York, New York 10018.
[4]American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.
[5]National Association of Corrosion Engineers, 1440 South Creek Drive, Houston, Texas 77084.

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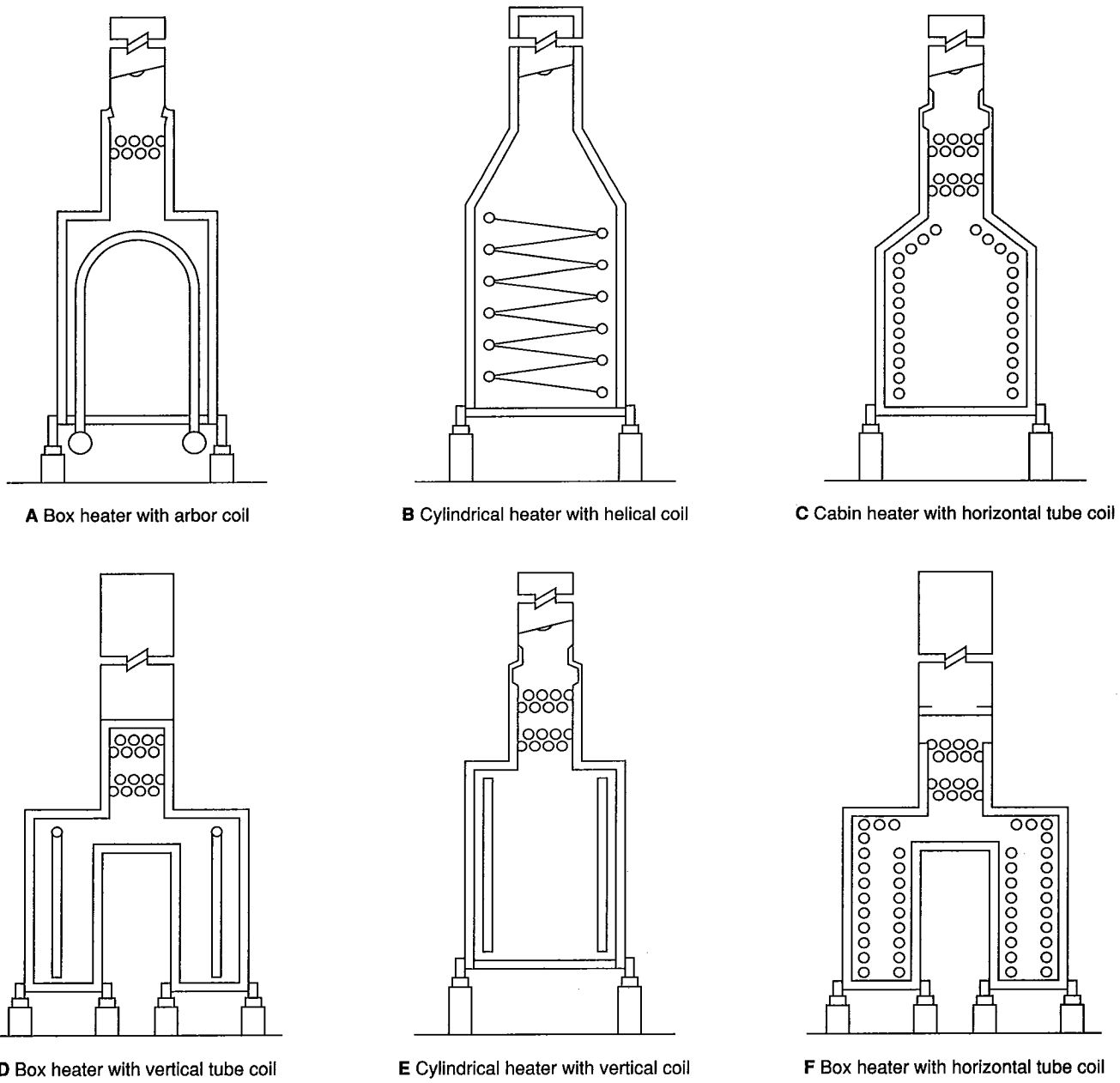


Figure 1—Typical Heater Types

Figure 2 shows a typical box-type heater with a horizontal coil and identifies the main heater components. This type of heater can have locations or zones of different heat densities.

The tubes in the radiant section of the furnace are called radiant tubes. The heat pickup in these tubes is mainly through direct radiation from the heating flame and the incandescent refractory.

The shock or shield tubes are normally located at the bottom of the convection section. Because these tubes absorb both radiant and convective heat, they usually receive the highest heat density.

The zone of lower heat density is the convection section. The tubes in this section are called convection tubes. The heat pickup in the convection section comes from the combustion gases, primarily through convection.

The size and arrangement of the tubes in a box-type heater are determined by the type of operation the heater is meant to perform—for example, crude oil distillation or crack-

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ing—the amount of heating surface required, and the flow rate through the tubes. Box-type heaters may be updraft or downdraft, with gas- or oil-fired burners located in the end or side wall, the floor, the roof, or any combination of these. After the oil convection tubes are installed in the convection section, auxiliary tubes are often installed to preheat air for the burners or to generate or superheat steam for process and other uses. In Figure 2 the convection section is centered in the upper portion of the box-type heater and the radiant tubes are on the two side walls.

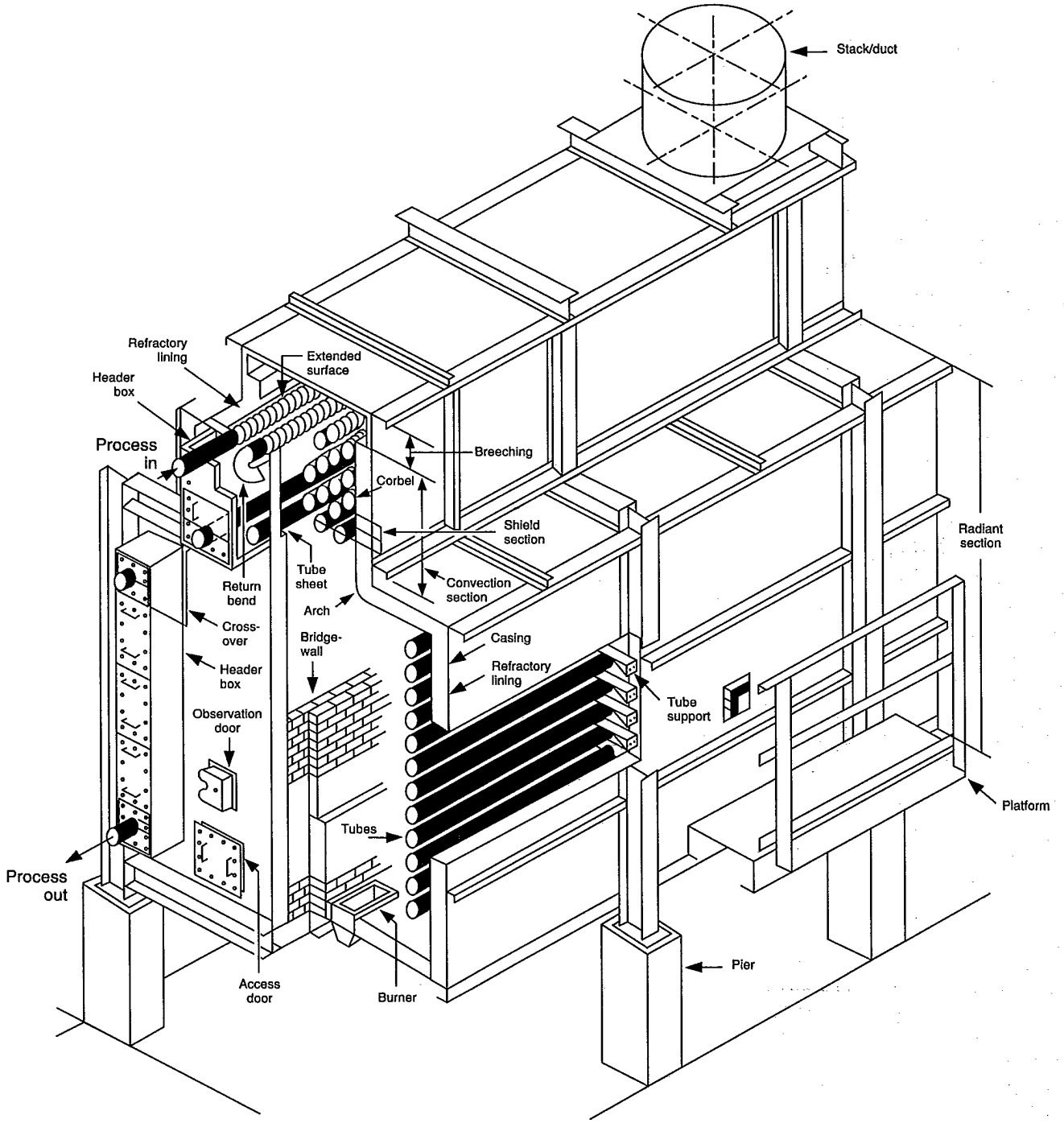


Figure 2—Box-Type Heater With Horizontal Tube Coil Showing Main Components

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1.3.1.3 Heaters With Vertical Coils

A vertical coil heater may be either cylindrical or rectangular (box type). The major portion of the heating coil consists of vertical tubes. In some installations, an oil-economizer (convection) section, an air-heating section, or both are used above the main vertical heating section. The tubes in an oil-economizer or air-heating section may be either horizontal or vertical. The purpose of an oil-economizer or air-heating section is to improve the economy of operation by increasing thermal efficiency.

Most vertical coil heaters are bottom fired, with the stack mounted directly on top of the heater. Downdraft vertical heaters have also been used.

1.3.1.4 Heaters With Helical Coils

Helical coil heaters are cylindrical with the surface of the radiant section in the form of a coil that spirals up the wall of the heater. They do not usually have a convection section, but if one is included, the convection surface may be in the form of a flat spiral or a bank of horizontal tubes. The stack of a helical coil heater is almost always mounted directly on top of the heater.

1.3.1.5 Heaters With Arbor Coils

Heaters with arbor or wicket coils are used extensively in catalytic reforming units for preheat and reheat service and as heaters for process air or gases. These heaters have a radiant section that consists of inlet and outlet headers connected with inverted or upright L or U tubes in parallel arrangement. The convection sections consist of conventional horizontal tube coils.

1.3.2 HEATERS REQUIRING SPECIAL PRECAUTIONS AND INSPECTIONS

1.3.2.1 Heaters in Hydrogen and Hydrogen Sulfide Service

Heaters used in hydrodesulfurization, hydroforming, hydrocracking, and similar processes often have austenitic stainless steel tubes. These installations usually process reactor feed or recycled gas, and the designs may be any of those discussed in the preceding subsections. The sulfide scale formed in these installations can react with water and oxygen to form polythionic acid (see API Recommended Practice 571). Precautions must therefore be taken during downtime to protect the tubes.

1.3.2.2 Heaters Used in Steam/Methane Reforming

Heaters used in steam/methane reforming—the vaporized feed may vary in content from methane to any light hydrocarbon—usually have many rows of parallel vertical tubes that operate from 1500°F (815°C) to 1800°F (980°C). Figure 3 shows a steam/methane-reforming heater. These heaters are normally down fired from the roof or side fired at many levels to achieve even heat distribution across the entire length of the radiant tubes. The tubes may be made of wrought high-strength materials, including the proprietary alloys Incoloy 800 and Incoloy 800H, or of cast materials, including HK40, HP, and their proprietary modifications. The connecting pipes between the tubes and the inlet and outlet headers are called *pigtails*. Outlet pigtails operate at a temperature of 1400°F (760°C) plus or minus 100°F (38°C) and must be designed with low stress levels, primarily in bending. To accomplish this, the tube support must be adequate, and tube bowing must be minimized.

When centrifugally cast tubes fail, the failure is generally due to stress rupture at the hottest, most highly stressed portion of the tube. The hottest areas are normally near the bottom or outlet of the tube, since the temperature of the gas inside the tubes rises during reaction by about 500°F (260°C), from about 900°F (480°C) to about 1400°F (760°C). If flame from burners or from combustion products deflected off walls impinges the tube, stress rupture can occur in the hottest parts of the tube.

Bowed tubes result from inadequate upper support or from heating on one side of the tube. The weight each upper support unit must bear varies from 50 to 100 percent of the

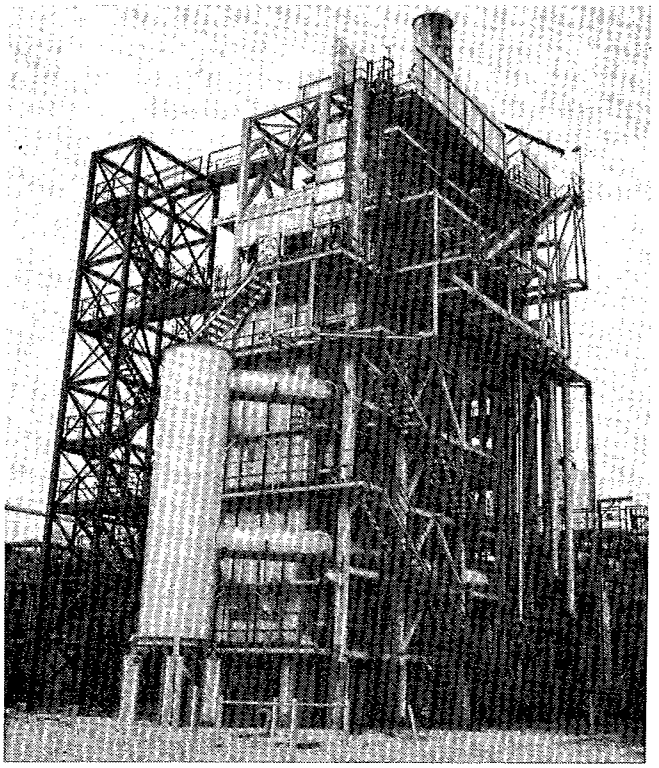


Figure 3—Steam/Methane-Reforming Heater

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total tube weight, depending on the design. Heating on one side of the tube causes greater thermal expansion on the hotter side and bowing toward the heat source. This continues until the restraints on each end of the tube prevent further bowing and cause higher stresses in the support assembly and the pigtails. Designs incorporating staggered tubes can minimize tube bowing with adequate uplift supports. Bent tubes have higher stress levels at their bends than do straight tubes.

Cast tube materials embrittle after exposure to high temperatures. Weld materials that embrittle during post-weld cooling have high residual stresses. Weld material with a carbon-silicon ratio that does not match that of the base metal fissures easily during welding. Any microfissures not detected during fabrication can propagate during subsequent heating, thermal cycles, or continual high stresses from bowing or localized heating. Welding flux must be removed from tube welds. Sandblasting is the recommended method of flux removal. Flux of lime with fluorides is corrosive if the combustion gases are reducing (because of very little excess air) and sulfur is present.

Centrifugally cast tubes fail in a stress rupture pattern that is different from that of most furnace tubes. A centrifugally cast tube usually has a thick wall. Thermal stresses are highest near the tube midwall. Stress rupture failures start as fissures in the midwall. As the on-stream time increases, the fissures progress to the inside diameter of the tube. The final stage of stress rupture occurs when the fissures reach the outside diameter.

High stresses on pigtails result in early failures. The failures usually occur at the inlet or outlet end of the pigtail. Bending stresses are a major cause of failure. These stresses result from tube bowing, tube movement, sagging of the pigtail under its own weight, thermal expansion of a pigtail loop, and any other condition that causes the pigtail to bend. Most pigtails are of Incoloy 800H or similar wrought materials, and most failures are cracks that develop from intergranular oxidation. If stresses are high, cracks will develop with little or no creep at start-up or shutdown. At lower stress levels, creep will occur, and intergranular oxidation will make the area slightly magnetic.

Furnace outlet headers have various designs. Those that are internally uninsulated have been made of cast materials conforming to ASTM A 297, Grade HT or HK, or of wrought materials, including Incoloy 800H. The cast headers have a history of cracking near any junctions, including inlets, outlets, laterals, tees, or elbows, because of embrittlement due to carbide precipitation and sigma formation. These headers are horizontal and do not float freely. The embrittlement that occurs does not allow any restraint of the thermal growth and results in high stresses with resultant cracking. Because of the embrittlement, welding repairs are difficult unless the surfaces are annealed or buttered with a ductile weld material before welding. In recent years, proprietary cast materials have been developed and used for furnace outlet headers and have had excellent service histories. These materials are not subject to the problems for HT and HK materials described in the preceding text.

Wrought headers operating at temperatures near 1400°F (760°C) have also had a good service history. They maintain ductility and can yield, by creep or stress relaxation, to reduce localized stresses. As in any high-temperature design, however, stresses must be kept low, particularly at supports and at openings in the headers. Any openings required in the headers should be circular or elliptical. If a square or rectangular opening is required, the corners must be cut on a radius to reduce concentrations of stress.

Some headers and outlet lines are made of carbon steel, C-Mo steel, or low–Cr-Mo steel and have refractory lining inside. Because the base metal is not resistant to hydrogen at high temperatures, the refractory must be sound to preserve its insulating properties. Refractory used in hydrogen and carbon monoxide service must have low iron and silicon content to avoid the possibility of hydrogen or carbon monoxide reacting with components of the refractory and the degradation of the refractory's essential properties. Start-up and shutdown procedures must minimize wetting of the refractory, partly to avoid destroying the insulating refractory and partly to avoid carbonic acid corrosion of the steel.

1.3.2.3 Pyrolysis Furnaces

Pyrolysis furnaces have many of the same problems that occur in steam/methane-reforming furnaces. The same materials are often used for both. There are a few major differences, however. Both Ys and U bends are used in pyrolysis furnaces and suffer erosion. The reaction in the tubes is usually carburizing and requires that the surfaces be smooth from boring or honing and that the material be more resistant to carburization. The material used in pyrolysis furnaces is often a modification of a high-strength material that is adequate in reforming heaters.

1.3.3 TYPES OF FIRED BOILERS

1.3.3.1 General

Fired boilers are boilers in which fuel is burned in a combustion chamber associated with the boiler. The heat of combustion is absorbed by the boiler to heat the water and convert it to steam. Fired boilers are either fire tube boilers or water tube boilers.

1.3.3.2 Fire Tube Boiler

A fire tube boiler consists of a drum with a tube sheet on each end in which the fire tubes are fastened. Water is contained within the drum surrounding the fire tubes. Fuel is burned in a combustion chamber associated with the boiler and arranged in such a manner that the heat and products of

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combustion (flue gases) pass through the inside of the fire tubes to heat the water surrounding them. The combustion products (flue gases) pass through the inside of the fire tubes to heat the water surrounding them. The combustion chamber may be a refractory-lined box located against one end of the drum or a steel chamber located within the drum and surrounded on all but one side by the water in the drum. In the first instance the boiler may be described as *externally fired;* in the second as *internally fired.*

Horizontal-return-tube boilers were popular in the early refineries. The Scotch marine boiler is of a fire tube design commonly employed in refinery package-type sulfur plants.

1.3.3.3 Water Tube Boiler

A water tube boiler usually has one or more, most likely two or three, drums and an external bank or banks of tubes connecting the two ends of the drum of a single-drum boiler or the two or more drums of multidrum boilers. In water tube boilers the water is contained within the drums and tubes. The fuel is burned in a combustion chamber arranged so that radiant heat and convection heat are transferred to the outside of the water tubes to heat the water within.

Water tube boilers may be either straight tube boilers or bent tube boilers. The tubes of most straight tube boilers are connected into headers, which in turn are connected to the boiler drums. Water tube boilers are always used when large steam capacities are needed. They are also used for high pressures and temperatures. They have been built in sizes up to 5,000,000 pounds (2,268,000 kilograms) of steam per hour, at pressures up to 5000 pounds per square inch gauge (34,474 kilopascals) and temperatures of approximately 1200°F (650°C).

Bent tube boilers are made in a variety of arrangements. They are similar to straight tube boilers, but they are almost always multidrum, and the tubes are connected directly into the boiler drums. The tubes are bent to allow them to enter the drums radially, to facilitate installation, to allow for expansion and contraction, and to allow for flexibility in design. Figures 4 and 5 illustrate typical bent tube boilers. Bent tube boilers may be either balanced draft boilers or positive pressure boilers.

Some boilers are fired using hot process waste gas streams, including fluid catalytic cracking unit (FCCU) regenerator flue gas as fuel to recover both sensible heat and fuel value. Carbon monoxide boilers are often used in refineries. Figure 6 illustrates one type of carbon monoxide boiler. Some refineries also use the combined cycle system, which utilizes the hot exhaust from gas turbines as combustion air in the boilers.

1.3.4 ECONOMIZERS AND AIR PREHEATERS

Economizers and air preheaters are heat exchangers that are used by some boilers as auxiliaries to recover more heat from the flue gases, heat that otherwise would be lost up the stack.

An economizer normally consists of a bank of tubes located in the path of the flue gases downstream of the steam-generating surfaces in the boiler. The low-temperature boiler feedwater is pumped through the tubes in this tube bank and is heated before passing into the boiler.

Air preheaters preheat the combustion air before it enters the combustion chamber. There are two basic types of air preheaters: *recuperative* and *regenerative.* The recuperative type is similar in principle to a conventional heat exchanger with the hot flue gases on one side of the heat transfer surface and the cool air on the other side. The most common recuperative type is the tubular air preheater, which consists of a tube bank with the tubes rolled into a stationary tube sheet at the top of the unit and a floating tube sheet at the bottom. This provides for difference in expansion caused by temperature differences between the tubes and the casing. In this type, the hot gases flow through the tubes, and the air passes around the tubes. Another type is made up of plates arranged with passages for the flue gas on one side of the plates and passages for air on the other side. Figure 7 illustrates a recuperative type of air preheater.

The most common regenerative type is called a *rotating heat transfer wheel* and is made up of many closely spaced sheets of metal. This metal absorbs heat as it rotates through the flue-gas compartment of its housing and gives up heat as it rotates through the air compartment (see Figure 8). The heat transfer wheel is rotated at approximately 3 revolutions per minute by driving a motor through a reduction gear. Diaphragms and seals divide the unit lengthwise to separate the hot flue gases from the air, which flow through the preheater in opposite directions.

The preheating of combustion air has high economic value. In the conventional air preheater, cold air from the forced-draft fan flows through the air preheater and extracts heat from the flue gases as they flow to the stack. Economizers or air preheaters are used when fuel savings justify them.

1.3.5 SUPERHEATERS

Superheaters consist of a bank of tubes located within the boiler setting, through which saturated steam flows from the steam drum and is superheated by the same flue gas that generates steam in the boiler. They may be of the radiant design, the convection design, or a combination of both, depending on the manner in which heat is transferred from the furnace gases to steam.

Superheaters may utilize tubes in hairpin loops connected in parallel to inlet and outlet headers. They may also be of the continuous tube design in which each element has tube loops in series between inlet and outlet headers. In either case, they may be designed for drainage of condensate or may be in pendent arrangements that are not drainable.

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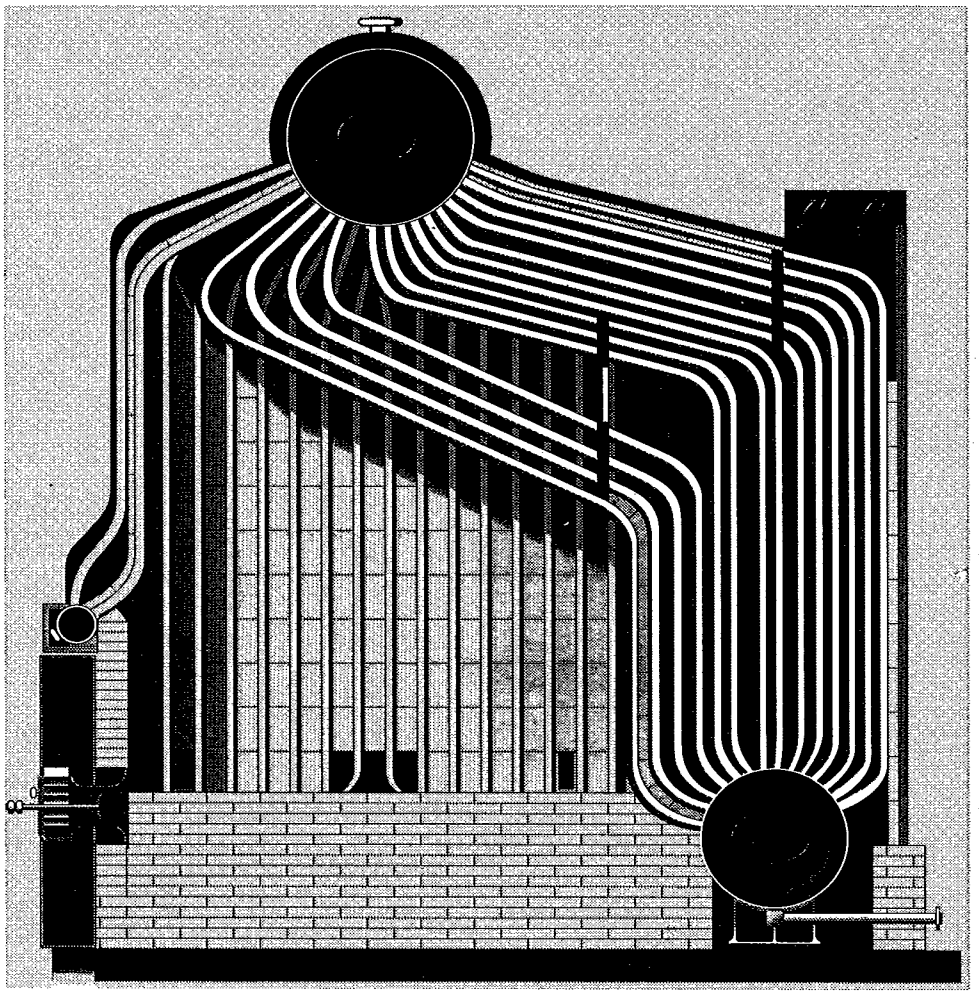


Figure 4—Typical Vertical Oil- or Gas-Fired Water Tube Boiler

Nondrainable or pendent arrangements are very susceptible to tube failure due to overheating on start-up. Water collected in the pendent must be slowly vaporized to assure a flow path for the steam. If the boiler is heated too rapidly, some pendents will not clear of liquid; therefore, steam will not flow and the tube will overheat and fail. Special start-up instructions should be taken into consideration with this type of arrangement. Both straight and pendent arrangement superheaters are susceptible to failure due to steam impurities. When steam is used in processing operations, superheated steam may be required to obtain the desired process temperature. Most of the large-capacity, high-pressure steam generators, especially those used for power production, are equipped with superheaters.

Superheated steam is also necessary for the most efficient production of power, especially when used in high-pressure, high-speed steam turbine drives.

SECTION 2—REASONS FOR INSPECTION

2.1 General

The reason for making the first inspection of a heater or boiler is to determine by comparison with the initial inspection at the time of construction or with basic records the effect that corrosion, erosion, and other factors have had on the equipment. The first inspection also helps to maintain the safety and efficiency of continued operation and forecasts maintenance and replacements, based on the indicated rate of deterioration.

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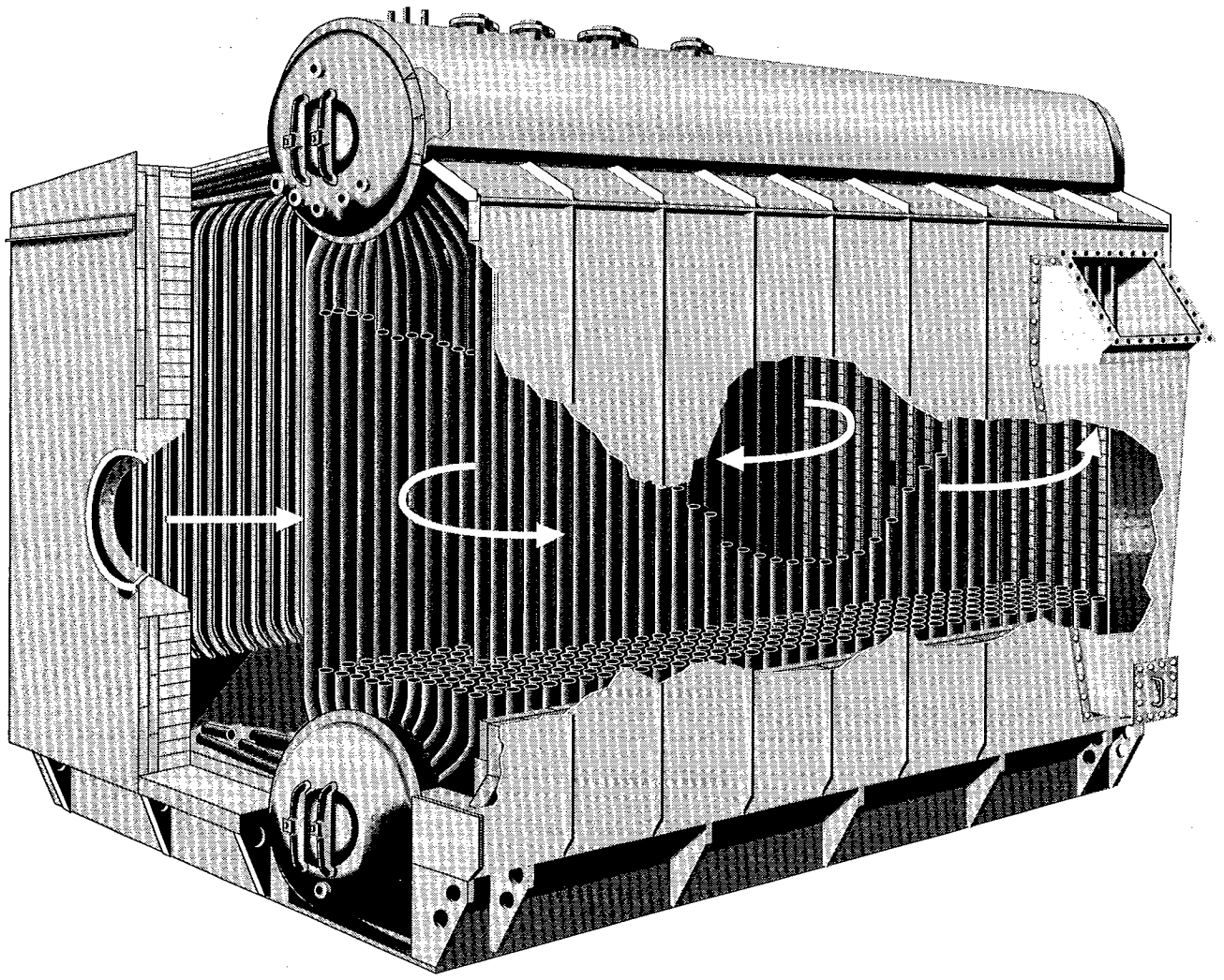


Figure 5—Another Variation of a Two-Drum Bent Tube Boiler

In the same way, all subsequent inspections are compared with the preceding inspection of the same specific purpose. The determination of the physical condition and the rates and causes of deterioration in the various parts makes it possible to schedule repairs or replacements before serious weakening or actual failure occurs. Many of the parts that make up a boiler or fired heater depend on some other part, and when deterioration and serious weakening occur in one part, some other part may become unprotected or overstressed. This can shorten service life.

It is possible to predict the repairs or replacements that will be required at the next scheduled down period by reviewing the data accumulated at regular inspections and by continual awareness of actual service conditions. If this information is available, all necessary drawings, lists of materials, and work schedules covering all phases of the work expected can be prepared. Necessary materials can then be estimated and replacement parts either wholly or partly fabricated at the most convenient times before shutdown. If work schedules are properly prepared and reviewed, each craft will know exactly what it has to do and the sequence in which the work should be done.

2.2 Inspection of Fired Boilers

The requirements governing inspection of boilers may differ widely from one jurisdiction to another. Under some laws, inspection must be made by state, municipal, or insurance company inspectors. Under other laws, inspections may be made by duly qualified plant inspectors. In either case, the inspector is usually commissioned by the regulatory authority and must submit reports of the inspection to the official

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responsible for enforcement of the boiler law. If the boiler is insured, inspection by the insurance company inspector will also serve to satisfy his company that the boiler is in an insurable condition.

Normally, governmental and insurance company inspectors will concern themselves only with the pressure parts of the boiler, the safety valves, level indicators, pressure gauges, and feedwater and steam piping between the boiler and the main stop valves, superheaters, and economizers. The refinery inspector must be concerned not only with this equipment but also with related nonpressure parts, including the furnace, burners, flue-gas ducts, stacks, and steam-drum internals.

2.3 Relations Between Outside Inspector and Plant Inspector

To reduce the length of boiler outages, joint inspections should be made by the outside inspector and the plant inspector. The outside inspector is primarily interested in seeing that minimum legal safety requirements are met. The plant inspector should be interested not only in safety but

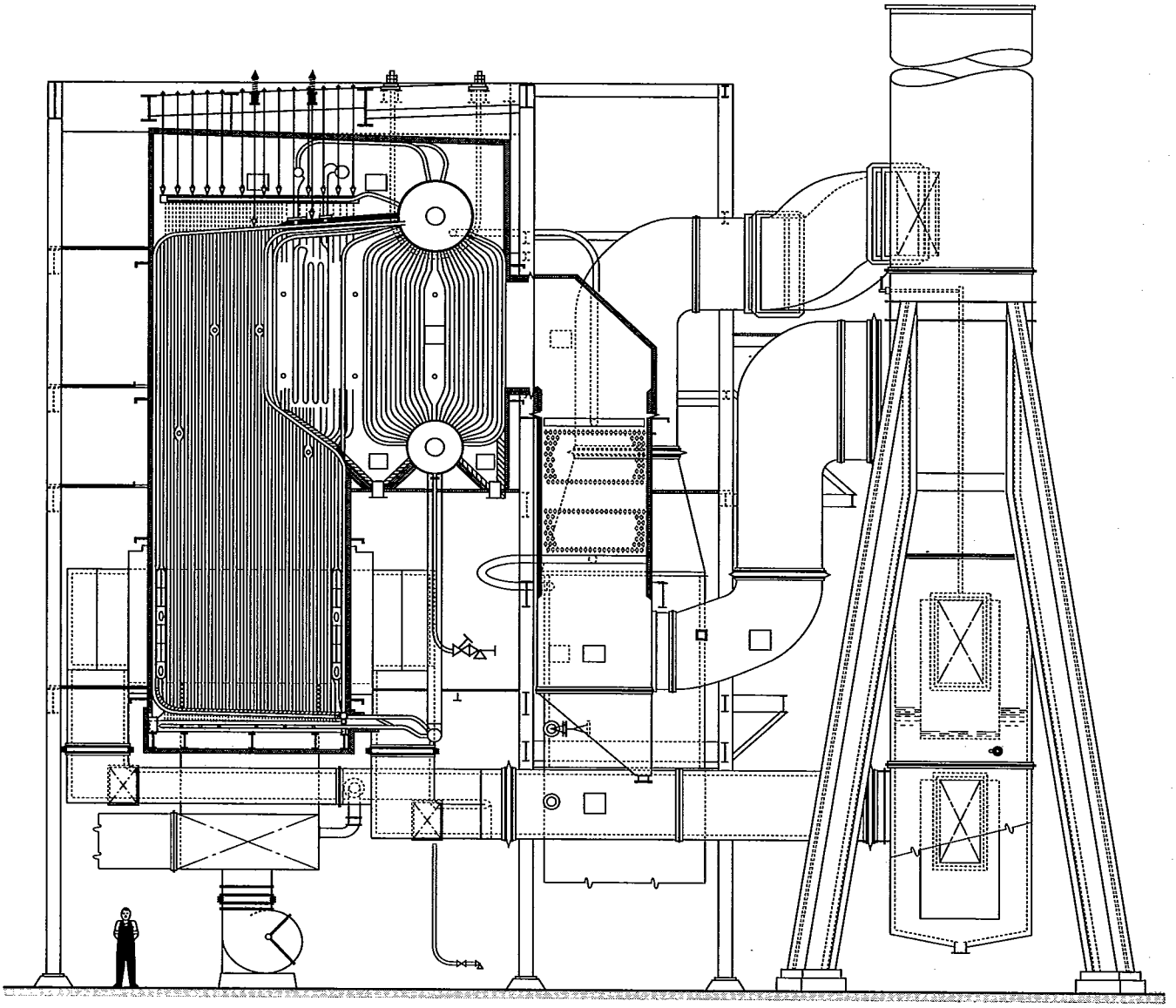


Figure 6—Typical Carbon Monoxide Boiler

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also in conditions that affect reliability and efficiency. Differences of opinion that might otherwise develop into troublesome problems can be resolved during a joint inspection. The outside inspector has an opportunity to examine many boilers that operate under widely varying conditions and often can offer valuable advice on the safe operation of boilers.

2.4 Factors Governing Frequency of Inspection

2.4.1 BOILERS

In most U.S. states and in some provinces of Canada, boiler inspection intervals may be set by law. In states or countries that have no such laws, the interval between inspections is set by the insurance carrier if the boilers are insured. In states or countries where there are no laws governing the construction and inspection of steam boilers, external and internal inspections should be scheduled periodically. Age of equipment, conditions of operation, type of equipment, kind of fuels, method of water treatment, and the like should be considered in determining the interval between inspections.

2.4.2 HEATERS

Heaters are usually part of a process unit, and the length of time that a heater may operate between inspections may be determined by other equipment. The shutdown of the entire unit makes the heater available for inspection. Advantage should be taken of every down period to inspect tubes, fittings, and the like, unless the operating time since the previous down period is too short to warrant such inspection.

Run lengths of heaters can be increased by on-stream inspection. This inspection can be visual, or temperature measurements of tubes can be made by using optical pyrometers, infrared techniques, and tube skin thermocouples.

API Standard 560 provides additional details to support inspection efforts at various sections.

SECTION 3—CAUSES OF DETERIORATION

3.1 Causes of Deterioration in Heaters

3.1.1 IN THE HEATING COIL

3.1.1.1 Type of Process

The main factor in the deterioration pattern of a heater is the operating process. The principal operating processes are crude oil distillation, vacuum distillation, asphalt or lubricating oil processing, hydrodesulfurization, cracking, reforming, light-distillate fractionation, and treating. The operating process determines the type of charge stock and is the main influence in establishing the basic operating conditions of the heater, which in turn influence the deterioration pattern.

3.1.1.2 Characteristics of the Charge Stock

The sulfur, chloride, organic acid, and solid material content are prime factors in determining the type and severity of deterioration. The sulfur content of the stock is an important factor because the type and rate of corrosion that can be expected on the internal surface of the heater tubes and fittings vary greatly with the sulfur, chloride, and organic acid content. Sulfur in particular is a determining factor in the choice of the material to be used to ensure satisfactory service life and maximum run length with a minimum of repairs or replacements. Hydrogen sulfide is a particularly corrosive compound whose corrosivity is usually increased when hydrogen is also present. Some charge stocks tend to produce deposits of coke or organic salts. Though not a direct cause of deterioration, these deposits can have a great influence on the temperature of the tube metal and can thus cause deterioration as a secondary effect. The removal of these deposits can also cause deterioration.

3.1.1.3 Velocity of Flow Through a Heating Coil

The velocity of flow through a heating coil may cause severe erosion in heater tubes and fittings if the velocity is critical or if direct impingement occurs. Erosion in heater tubes is usually the result of velocity. Erosion in heater fittings usually results from a combination of impingement and velocity. If the charge rate on a heater is materially increased, the increased velocity may cause metal loss from erosion and corrosion.

3.1.1.4 Pressure

An operating pressure that is permissible for the operating metal temperature is not a cause of deterioration when the metal temperature is below the creep range. When the operating metal temperature is above the temperature at which creep takes place, a slow stretching of the metal occurs, which may cause rupture after a long operating period. Excessive pressure may cause rapid creep of the metal and may result in bulging, cracking, and even complete failure by stress rupture in a comparatively short operating period. Creep and stress rupture are evaluated in API Recommended Practice 571.

3.1.1.5 Temperature

The operating temperature of a process heater is a factor in establishing the metal temperature of the tubes and fit-

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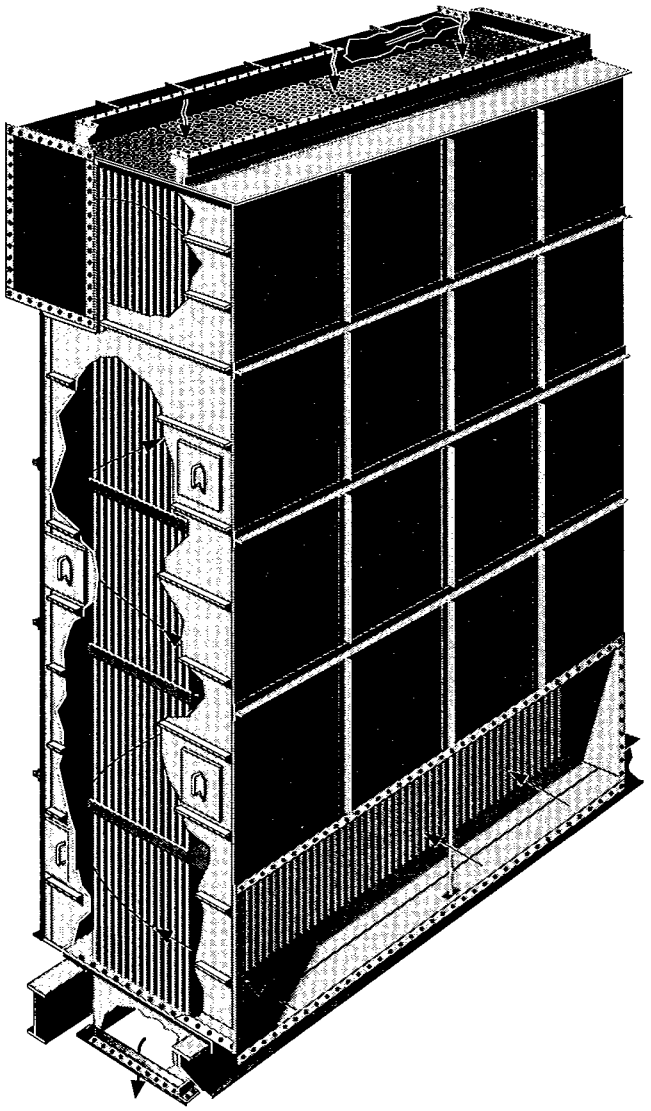


Figure 7—Tubular Air Preheater (Recuperative Type)

tings. The metal temperature plays a major role in the type and severity of the deterioration of the heater tubes. The metal temperature of individual tubes or along the length of any specific radiant tube of a given heater can vary considerably. The principal causes of abnormal variation in metal temperature are fouling of the tubes and improper or poor firing conditions in the heater.

The following are the types of tube deterioration associated with high metal temperatures:

a. Sagging. This is usually due to a decrease in the structural strength of the tube caused by overheating. It may also be caused by improper spacing of hangers, uneven metal temperatures, or failure of one or more tube supports or hangers.

b. Bowing. This is generally caused by uneven metal temperatures, which may be due to flame impingement or coke accumulation inside the tube. It may also be caused by binding of the tube in the tube sheets or improper suspension of the tube so that longitudinal expansion is restricted or by the use of improper tube lengths when individual tube replacements are made.

c. Oxidation or scaling. This may either be a localized condition or extend over the entire length of the tube inside the heater. Oxidation or scaling is usually the result of fouling of the tubes or general overfiring of the heater, which raises the metal surface temperature to a level at which oxidation occurs. Combustion deposits may have the appearance of oxide scale, but they can be distinguished by checking them with a magnet. Oxide scale is magnetic, and combustion deposits are not.

d. Bulging or creep leading to rupture. The strength of metal is reduced by high temperatures, and the stress for long periods of time will cause the hottest tubes to deform or creep. Creep failures can be avoided by using stress values based on the high-temperature properties of the metal. Bulging results when local overheating of the tube raises the metal temperature above the point at which the tube can withstand the stress due to pressure.

e. Metallurgical change. Steels subjected to high temperatures and stress for long periods undergo metallurgical change. This change results in various conditions, including carburization, decarburization, and grain growth. All of these conditions lead to a general reduction in mechanical strength or a change in ductility, which may eventually result in complete failure of the material. Some materials, including 5Cr-0.5Mo-P and other trace elements above 0.015 percent may be subject to precipitation hardening after exposure to furnace operating temperatures for a sufficient period of time. The result may be temper embrittlement with a loss of elongation and notch ductility as these elements precipitate to the grain boundaries after about a year at temperatures from 572°F (300°C) to 1112°F (600°C). Accordingly, brittle cracking at transition temperatures as high as 300°F (149°C) have been encountered. See API Recommended Practice 571 for a detailed description of these forms of deterioration.

f. Effects of expansion. All metals expand when heated. Elevated temperatures cause expansion that, if not properly provided for, will result in stresses that are sufficient to cause serious weakening and deformation of the tube or fitting.

g. Increased corrosion. Internal and external corrosion are strongly influenced by temperature. Differences in the corrosion rate along the length or around a cross section of a tube are often the result of temperature differences.

h. Thermal fatigue. Metal that operates under cyclic temperature conditions, especially over a wide range, may develop cracks because of thermal fatigue. Cracks start at the surface of the material, progressing slowly at first and then more rapidly with each cycle of temperature change. Thermal fa-

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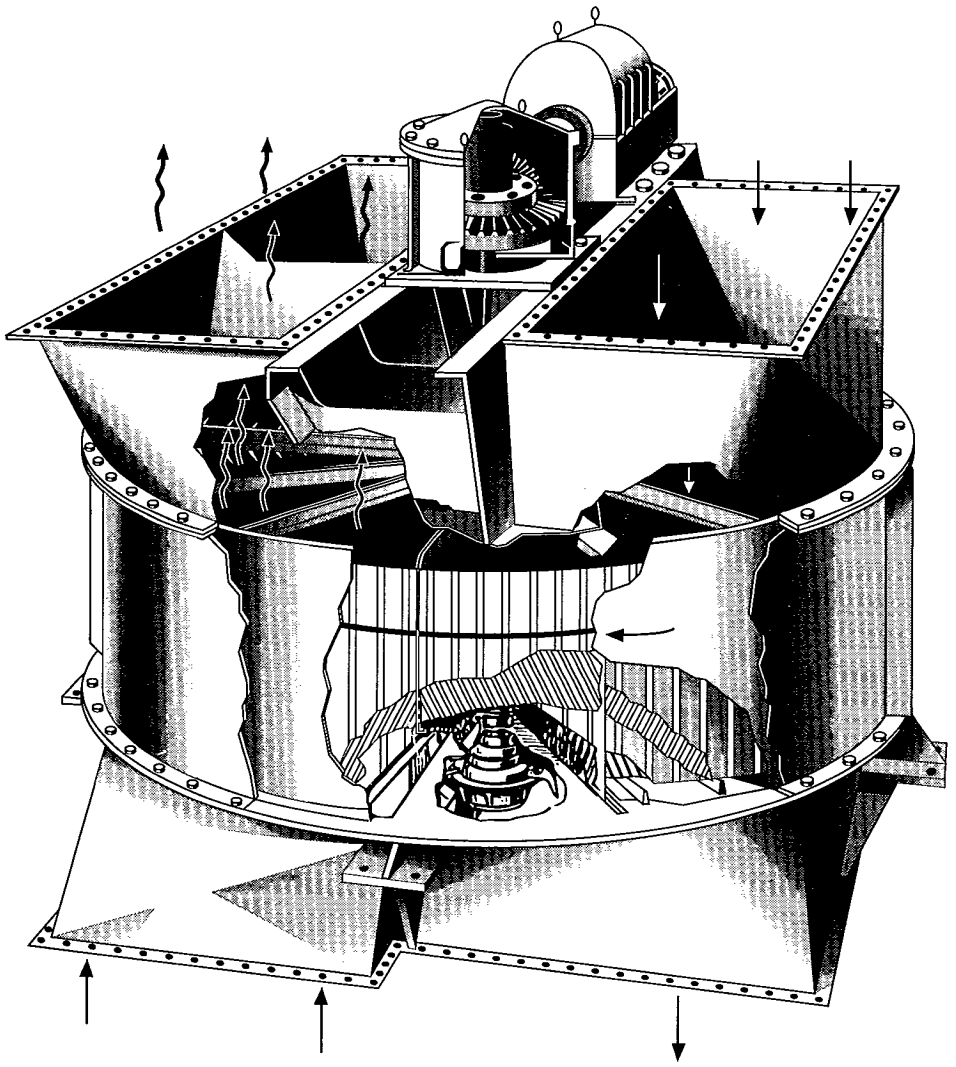


Figure 8—Regenerative Air Preheater

tigue is often found at locations where metals that have different coefficients of expansion are joined by welding.

i. Thermal shock. This is caused by a sudden marked change in temperature either from hot to cold or from cold to hot. The stresses resulting from the sudden unequal expansion or contraction of the different parts may cause distortion only or distortion plus cracking. Thick metals are more susceptible to cracking than are thin ones. The most likely time of temperature shock is during unit start-ups. Heating or cooling rates should be controlled to avoid thermal shock.

3.1.1.6 Combustion Products

The corrosion problems that result from the combustion of furnace fuels depend primarily on the character of the fuel. When the gas or fuel oil has a high sulfur content, one of the combustion products formed and deposited on the outside surfaces of the tubes is a sulfate. This sulfate is harmless during periods of operation, but as soon as the deposit is allowed to cool it becomes highly hygroscopic and takes up moisture from the air, hydrolyzing to produce sulfuric acid, which immediately attacks all metal with which it is in contact.

When the fuel has a high vanadium content, metal at temperatures above a critical point in the range from 1200°F (650°C) to 1400°F (760°C) is subject to very rapid attack from vanadium pentoxide. The vanadium pentoxide deposits on the hot metal surface and causes fluxing and melting. After a certain amount of deposit has accumulated, it sloughs off and the attack cycle starts again.

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Convection sections where flue-gas dew-point temperatures occur during operations suffer metal loss because of acid material from the products of combustion. Metal loss on the exterior of convection tubes may be difficult to evaluate because of inaccessibility.

3.1.1.7 Mechanical Deterioration

Mechanical deterioration may materially reduce the service life of heater tubes and fittings. The two most common causes of this are leakage in the tube rolls—the rolled joints between tubes and fittings—and damage during mechanical cleaning.

Leakage in the tube rolls presents a mechanical difficulty. It may result from faulty procedures or workmanship when the tubes were originally installed or it may be caused by thermal upsets during operation.

Damage to a tube during mechanical cleaning may be caused by faulty procedures or workmanship. One of the most common causes is allowing the cleaner to operate in one position for so long that it cuts the tube metal. Machined surfaces of plug-type header fittings can be damaged by contact with cleaning tools.

Undue force used by workers in closing fittings may result in the development of cracks in the fitting body or at the base of fitting ears and may cause excessive wear or distortion of the plugs of U-bend seats, fitting ears, or holding sections and members—dogs or caps and screws. The use of excess force commonly occurs because of improper cleaning of ground surfaces or mismating of plugs to return bends. Training and close supervision of workers with regard to the proper care, use, and amount of tightening permissible are essential to prevent this damage. Casting or forging defects may also result in cracks in the fitting body or at the base of fitting ears.

It is common practice to heat fittings to aid the removal of plugs and to reduce the chance of damaging the casting. Overheating with a torch may cause the fitting to crack. The depth and seriousness of cracks formed by overheating with a torch should be investigated.

Steam/air decoking can cause serious oxidation and other deterioration of tubes unless temperatures are carefully controlled.

3.1.2 IN THE SETTING

3.1.2.1 Climatic Conditions

The rate of deterioration caused by climatic conditions primarily depends on whether the atmosphere is dry, humid, or salty and on the industrial fumes that may be present. Deterioration resulting from a humid atmosphere may not be due to geographic location but may be the result of the location of the heater within the refinery. Location near cooling ponds or towers when the prevailing winds are toward the heater may cause deterioration.

The types of deterioration that result from climatic conditions are rusting of exposed or unpainted steelwork, general deterioration of painted surfaces, and erosion and further deterioration of the external housing of a heater. If the external housing is allowed to deteriorate, rain or other moisture will enter the openings and deteriorate the internal refractory, insulation, and steelwork, especially while the heater is out of service for any reason (see API Recommended Practice 571).

3.1.2.2 Operating Temperatures

Firing conditions and furnace temperature are the main causes of deterioration of the materials that form the internal lining of the heater. The severity of the deterioration will vary with the furnace temperature, which in turn is determined by the process operating conditions.

The purpose of the internal materials, including refractory or insulating linings, is to provide protection from heat to the structural steel framing, roof structures, and tube sheets and to improve the thermal efficiency of the heater. At high temperatures, refractory will deteriorate after long-term exposure by spalling, failure of the binding material, melting, and loss of structural strength. When the insulating value of refractory or insulating material is reduced, the supporting steel is subjected to high temperatures and may deteriorate rapidly as a result of oxidation, scaling, and possible metallurgical changes.

3.1.2.3 Products of Combustion

Corrosive agents are produced in the combustion of fuels that contain sulfur and vanadium. The types of corrosion that can result from the burning of fuels with high sulfur and vanadium content are mentioned in 3.1.1.6.

Deterioration due to sulfur will occur on cold steelwork when it has been exposed to the furnace gases as a result of deterioration of the refractory or insulating linings or if a furnace is operated under a positive pressure. It is imperative that the outer casing of furnaces be maintained in a tight condition. When flue gases are permitted to permeate to the atmosphere at various locations, they deposit sulfur on the casing and metal parts that are below the dew point. Such deposits form acids, accelerating corrosion of the casing and the refractory supports.

When fuel ash and refractory are in contact at a moderately high temperature, fluxing may occur, producing a slag that may be fluid. Metal oxides, including those of vanadium, molybdenum, and sodium, are fluxing agents. At least three deteriorating actions of this slag formation can be recognized:

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a. Melting.
b. Penetration.
c. Chemical action.

The general effects of this slagging are to decrease the thickness and reduce the insulating effect of the refractory and thereby to allow a high metal temperature on the supporting steel parts.

3.2 Causes of Deterioration in Fired Boilers

3.2.1 OVERHEATING

Overheating is one of the most serious causes of deterioration of boilers. Overheating of the boiler tubes and other pressure parts may result in oxidation, accelerated corrosion, or failure due to stress rupture. Overheating develops from abnormal conditions, including loss of coolant flow or excessive boiler gas temperatures. These abnormal conditions may be caused by inherently faulty circulation or obstructed circulation resulting from water tubes partly or wholly plugged by sludge or dislodged scale particles. Overfiring or uneven firing of boiler burners may cause flame impingement, short-term overheating, and subsequent tube failure. The results may be oxidation of the metal, deformation of the pressure parts, and rupture of the parts, allowing steam and water to escape. Figures 9 and 10 show boiler tubes that have failed because of overheating.

Boiler tubes may be damaged by poor circulation. Under certain conditions of load and circulation, a tube can become steam-bound long enough to overheat locally and fail. If circulation is periodically reestablished, the hot portion of the tube is quenched by relatively cool water. This often causes thermal fatigue cracks, which may eventually result in tube failure. This condition can also result in caustic or chelate corrosion.

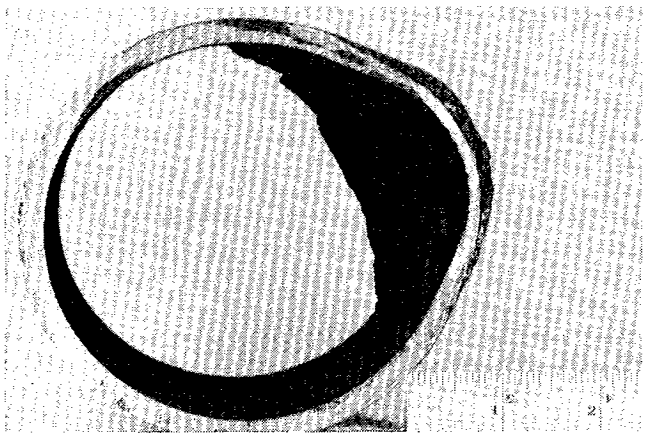


Figure 9—Short-Term Boiler Tube Failure Caused by Waterside Deposits, Subsequent Overheating, and Final Bulging of the Tube Wall

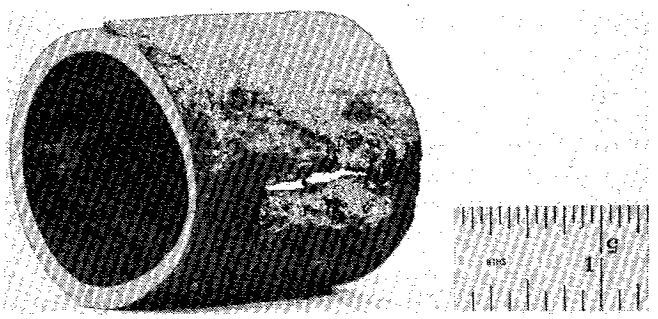


Figure 10—Longer Term Boiler Tube Failure Caused by Poor Circulation and Subsequent Overheating, Oxidation, and Final Failure by Stress Rupture

Steam binding may be caused by the insulating effect of slag deposits on the outside of the lower part of the tube. This demonstrates the importance of avoiding, as much as possible, nonuniform slagging of waterwalls. Steam superheaters can become overheated and severely damaged during start-up if cold boilers are fired at an excessive rate before a sufficient flow of steam is established to keep the superheaters cool. They can also become overheated if the steam vented from the superheater outlet is not sufficient to provide steam flow through the superheater during warmup or low-load operations. The overheating results in warped tubes and oxidation of the tube metal, leading to early tube failure.

The faulty operation of steam-separating devices may result in deposition of boiler water solids in the superheater tubes, with subsequent damage to the tubes from overheating.

Nonpressure parts, including refractory linings of furnaces, burners, supporting structures, and casings, may also be damaged from overheating. Usually, such overheating is caused by improper operating conditions or is a result of deterioration of other protective parts. For example, if the refractory lining of a furnace is permitted to deteriorate from normal wear, erosion, spalling, or mechanical damage, it will no longer protect the outer furnace casing and structural supports adequately, and such parts may in turn begin to deteriorate rapidly.

3.2.2 CORROSION

3.2.2.1 General

Corrosion may occur on all external and internal surfaces of boiler parts, economizers, and air preheaters. The extent and rate of deterioration caused by corrosion will depend on the condition of the feedwater, the type and quality of fuel burned, the quantity of excess air utilized in combustion, and

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the prevailing atmospheric conditions. Frequency of start-ups and shutdowns also affects the rate of this deterioration.

3.2.2.2 Waterside Corrosion

Corrosion of tubes and other internal surfaces is largely dependent on the water and water chemistry used within the boiler. Some of the more common types of waterside corrosion include caustic corrosion, dilute acid corrosion, pitting or localized corrosion, and stress corrosion cracking. A significant factor in the degree of waterside corrosion is the amount of corrosion product deposited. Deposits restrict the heat transfer and lead to overheating. Depending on which contaminants are present in the feedwater during the period of chemical unbalance, different deposition locations, rates, and effects will be experienced.

Caustic corrosion, sometimes called caustic gouging, develops from deposition of feedwater-corrosion products in which sodium hydroxide can concentrate to high pH levels. At high pH levels, the steel's protective oxide layer is soluble and rapid corrosion can occur. Deposits normally occur where flow is disrupted, in areas of high heat input. When the deposit thickness is great enough to make caustic concentrations locally corrosive, severe corrosion resulting in irregular thinning or gouging of the tube wall can occur. Figure 11 illustrates this form of localized corrosion.

Hydrogen damage may occur if the boiler is operated with low-pH water, which may be caused by the ingress of acidic chemicals from the water treatment facility, a leak in a saline-cooling water condenser, contamination from chemical cleaning, or other factors that may lower the boiler feedwater pH to less than 7. Close control over boiler water chemistry and monitoring practices are important factors in preventing hydrogen damage.

Boiler tube failures caused by pitting or localized corrosion result from oxygen attack on the internal side of the boiler tube. Pitting corrosion of economizer tubing normally results from inadequate oxygen control of the boiler feedwater. For full protection against oxygen pitting during shutdown, the boiler should be kept full of hydrazine-treated water and blanked or capped with nitrogen. Figure 12 illustrates a boiler tube with a through-wall oxygen pit.

While stress corrosion is usually associated with boilers in which austenitic tubes are used for superheater and reheater tubing, failures have occurred in ferritic tubes where a desuperheater or attemperator spraying station introduced high levels of caustic concentration. Stress corrosion cracking of B-7 studs may also occur in areas where a leaking gasketed joint may allow caustic concentration.

3.2.2.3 Fireside Corrosion

Fuel constituents and metal temperatures are important factors in the promotion of fireside corrosion. Two main kinds of fireside corrosion are associated with low-temperature attack and high-temperature oil-ash corrosion.

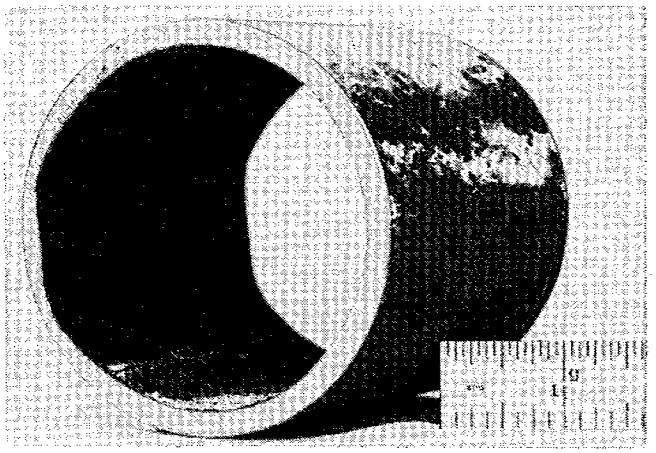


Figure 11—Uneven Corrosion of the Tube Wall Caused by Caustic Gouging

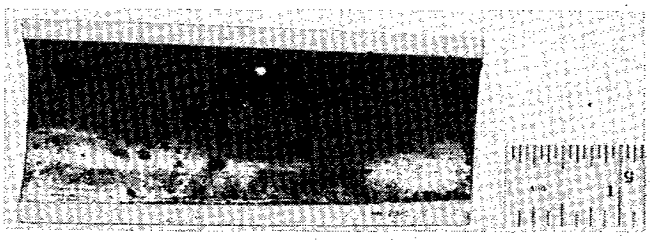


Figure 12—Boiler Tube Showing Penetration of the Tube Wall by a Localized Oxygen Pit

Corrosion may occur on the flue-gas side of economizer and air preheater tubes. The severity of this corrosion depends on the amount of sulfur oxides or acid in the fuel burned and on the temperature of the flue gas and of the media being heated. When sulfur oxides are present in the flue gases, corrosion tends to be severe if the gases cool down to the dew-point temperature. The gas temperature in economizers and preheaters must be kept above 325°F (163°C) to prevent condensation of corrosive liquid. This may be best effected by designing the tubing and the water flow in the tubing so that the gas temperatures are controlled as noted in the preceding text.

External corrosion of boiler parts may be expected when boilers are out of service for long periods of time. The sulfurous acid formed from the reaction of condensed moisture with the sulfur in ash deposits can cause rapid corrosion of boiler parts. Also, if a unit remains idle for an appreciable length of time, a warm humid atmosphere tends to corrode

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boiler parts and supports, unless adequate mothballing procedures are followed.

Fuel-ash corrosion may occur when fuel ash and refractory are in contact at a moderately high temperature. Fluxing occurs and produces a slag that may be fluid. Metal oxides, including those of vanadium, molybdenum, sodium, and sulfur, are fluxing agents. At least three deteriorating actions of this slag formation on refractory and boiler metal parts can be recognized:

a. Fluxing and melting.
b. Penetration.
c. Chemical action.

The general effect of this slagging is to decrease the thickness and reduce the insulating effect of the refractory, thereby causing a high temperature of the protected steel parts. The slagging effects of vanadium and sodium oxides may also cause rapid deterioration of boiler hardware, including tube hangers and spacers. The use of fuel-oil additives or a change in metallurgy to 50Cr-50Ni or 60Cr-40Ni alloy reduces the effects of this type of corrosion. Some designs incorporate steam-cooled spacers and hangers, which control this form of corrosion.

3.2.3 OTHER FORMS OF DETERIORATION

Mechanical deterioration of boiler parts can result from a number of causes:

a. Fatigue from repeated expansion and contraction and corrosion-fatigue from the combined action of fatigue and corrosion.
b. Abnormal stresses imposed by rapid changes in temperature and pressure, especially in the case of thick-walled drums.
c. Improper use of cleaning tools.
d. Improper use of tube rollers.
e. Settlement of foundations.
f. Excessive external loadings from connected piping, wind, earthquake, and similar sources.
g. Breakage and wear of mechanical parts.
h. Firebox explosion.
i. Vibration due to improper design or support failure.
j. Improper gaskets that allow steam leaks to score the seating surface.
k. Nonweathertight casing that allows external tube corrosion during extended shutdowns.

If metal is repeatedly stretched, compressed, bent and straightened, or otherwise worked, it will eventually become fatigued and brittle and may crack under a stress far below its normal breaking load, as discussed in API Recommended Practice 571. Because of temperature changes involved in putting a boiler out of service and back into service and the normal temperature fluctuations during operation, the metal in boiler parts may become fatigued from expansion and contraction. Tubes may also become fatigued as a result of alternate wetting by steam and water, which causes fluctuating conditions. If corrosion acts in conjunction with fatigue, the fatigue resistance of the metal will be reduced because of the corrosive medium, and corrosion-fatigue cracks will result. Corrosion-fatigue cracks have been found with welds in deaerators.

When very rapid temperature changes occur in metal parts, especially thick metal parts, the parts may be overstressed by the expansion or contraction of the portions of the metal that have changed in temperature against the portions of the metal that have not changed in temperature. A similar situation exists when a glass tumbler is only partly filled with hot liquid and shatters.

Tube cleaners improperly employed—allowed to operate too long in one position, for example—may cause damage by cutting grooves inside the tube.

Improper use of tube-rolling tools by underrolling or overrolling may cause tube-roll leaks or damage to the tube ends or tube seats.

Foundation settlement may be a serious cause of deterioration in boilers because of the severe stresses that may be set up in the complicated interconnection of parts, in the external piping, and especially in the refractory linings and baffling. Excessive loads on the boiler by the connection of large pipe lines may cause damage to the boiler foundation and pressure parts.

Settlement of foundations may also result from heat transmission from the firebox and subsequent drying of the soil.

In earthquake zones, earthquakes may cause severe damage. The damage will be somewhat similar to that caused by foundation settlement and may be particularly severe to refractory linings.

Vibrations from high and moderate winds, earthquakes, burner operating instability, and high flue-gas flow across tube banks can cause damage to various parts of boilers as follows:

a. Stacks may be so damaged that they overturn.
b. Air and flue-gas ductwork may be damaged, resulting in cracks at corners or connections.
c. Expansion joints may crack.
d. Guy lines may loosen or break.
e. Piping and tubing may be overstressed and fail.
f. Anchor bolts of stacks may be overstressed and fail.

Breakage and wear of mechanical parts are probably the most common forms of deterioration of the various parts and auxiliaries of boilers, especially burners and equipment that handles solid fuel and ash. The associated services are very severe, involving high temperatures, almost continuous operation, and extremely abrasive operating conditions when solid fuels are used.

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SECTION 4—SAFETY PRECAUTIONS, PREPARATORY WORK, AND CLEANING

4.1 Safety

Safety precautions must be taken before any heater, boiler, flue duct, or stack is entered. In general, these precautions consist of adequate ventilation to remove all flue gases, reduction of temperature to a safe level for personnel, blanking or disconnection of all lines, and when two or more heaters or boilers are connected to one stack, blanking of the flue duct to prevent the entry of flue gases from other active units. Isolation from all other piping and equipment should be established.

Dust and acid-containing material on internal surfaces are to be expected. The problem they present may be complicated if fuel-oil additives that leave toxic residues have been used. Protective equipment must be made available and used until it has been determined that safe conditions exist. When vanadium dust is present, protective apparatus and clothing must be used when internal inspections are performed.

Note: Consult all applicable OSHA and other federal, local, and state safety rules and regulations.

4.2 General Preparatory Work

Before the inspection, the tools needed for inspection should be checked for availability, proper working condition, and accuracy. This includes tools and equipment that are needed for personnel safety. Safety signs should be provided where needed before work is started. The following tools are needed to inspect fired heaters and stacks:

a. Portable lights, including a flashlight.
b. Thin-bladed knife or scraper.
c. Broad chisel or scraper.
d. Pointed scraper.
e. Inspector's hammer.
f. Inside calipers.
g. Outside calipers.
h. Direct-reading calipers or special shapes.
i. Mechanical tube caliper or micrometer for measuring the inside diameter of tubes.
j. Pocketknife.
k. Steel rule.
l. Special D calipers.
m. Pit depth cage.
n. Paint or crayon (see Note 2 below).
o. Notebook.
p. Magnifying glass.
q. Wire brush.
r. Plumb bob and line.
s. At least one type of special thickness measurement equipment (see next list).
t. Small mirror.

Note 1: No aluminum tools, including gauges or ladders, should be allowed near furnace tubes made of high-temperature alloy—for example, HT or HK alloy.
Note 2: Paint used on austenitic stainless steel should be chloride free and should not contain aluminum, zinc, lead, or sulfur, which could penetrate and damage metal. Yellow keel should not be used because of its sulfur content.

The following tools should be readily available in case they are needed:

a. Surveyor's level.
b. Carpenter's or plumber's level.
c. Magnetic-particle inspection equipment.
d. Liquid-penetrant inspection equipment (see Note 3 below).
e. Radiographic inspection equipment.
f. Ultrasonic inspection equipment.
g. Megger ground tester.
h. Sandblasting equipment.
i. Micrometer (0–1 inch).
j. Electronic strain gauge caliper.
k. Borescope.
l. Fiberscope.

Note 3: The liquid penetrant should be chloride and sulfur free.

Other related equipment that may be provided for inspection includes planking, scaffold material, a bosun's chair, and portable ladders. If external scaffolding is required, it may be possible to erect it before the unit is shut down.

Before the inspection is started, all persons working around a fired heater or boiler, flue duct, or stack should be informed that people will be working on the inside. A safety guard should be stationed at the inspection door of the equipment being inspected. This person can serve as a guard and can also record data from the inspection findings.

Personnel working inside this equipment should be informed when any work is going to be done on the outside so that any unexpected noise will not cause needless alarm. Vibration of the tubes and the setting should be minimized while internal inspection work is being performed to prevent injuries due to the dislodging of loose refractory.

4.3 Preparatory Work Before Blinding or Opening Stainless Steel Tubes in Hydrogen and Hydrogen Sulfide Service

A thin film of iron sulfide forms on stainless steel tubes in hydrogen and hydrogen sulfide service. When this scale is exposed to moisture and oxygen it can hydrolyze to form polythionic acids. The polythionic acids can cause intergranular cracking of sensitized stainless steels. All grades of stainless steel that operate at temperatures above 650°F (343°C) will eventually sensitize.

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Note: Some sensitization resistance may be achieved through the use of stabilized and low-carbon grades of stainless steel.

There are two ways to avoid polythionic acid cracking. The more common one is to keep the tubes pressurized with inert gas. When blinding is required, a positive flow of inert gas should be maintained while the flanges are open and a blind is being installed. If desired, a small amount of ammonia can be added to the inert gas as a neutralizing agent. Maintaining a positive flow of inert gas excludes air and moisture. When tubes, crossovers, headers, or other parts of the furnace must be opened, a soda-ash wash is used. The usual solution is 2 weight percent soda ash (Na_2CO_3) with a suitable wetting agent. The solution should be circulated so that all gas pockets are moved and all surfaces are wetted. Sodium nitrate at 0.5 weight percent may also be added to the solution to inhibit chloride cracking. The solution may then be drained and reused in piping or another furnace. The 2-percent solution contains enough soda ash to leave a film, but a weaker solution may not. The film is alkaline and can neutralize any reaction of iron sulfide, air, and water. It is important to remember that the film, the residue from the soda-ash solutions, must not be washed off during downtime. Most units are put back on stream with the film remaining. If the film must be removed, flushing during start-up followed by inert gas may be acceptable. NACE RP01 covers this subject and the procedures involved.

4.4 Cleaning

4.4.1 EXTERNAL CLEANING

Tubes may be externally cleaned by various methods. The specific method is usually determined by the accessibility of the tubes and the purpose for which they are to be cleaned. Tubes that are readily accessible may be cleaned by wire brushing or sandblasting. Sandblasting is preferred if defects are suspected and a close inspection is required, since all deposits can be removed and the bare metal exposed. Refractory should be protected from sandblasting.

Because of tube arrangement, it is usually physically impossible to clean the economizer or convection tubes by wire brushing or sandblasting. Other methods, including the use of a steam lance or a stream from a water hose or high-pressure water equipment, may be used. In such instances, cleaning is performed primarily to remove external deposits and improve the heat transfer. Before resorting to steam or water cleaning of the tubes, careful consideration should be given to possible damage to the refractory insulation and brickwork, particularly in a service where a fuel with a high sulfur content is used.

4.4.2 INTERNAL CLEANING—HEATERS

The internal cleaning of tubes and fittings is usually required when fouling or coking is indicated by an increased coil pressure drop, an increased firing rate to maintain the desired coil outlet temperature, a decreased coil outlet temperature, or tube hot spots. Internal cleaning may be accomplished by several methods.

One method of cleaning is to circulate gas oil through the coil after the heater has been shut down and before the coils are steamed and water washed, before the start of inspection work. This method is effective if the type of deposits in the coil will be softened or dissolved by the gas oil. When tubes are coked or contain a hard deposit, other methods may be used, including steam/air decoking, hydroblasting, mechanical cleaning for coke deposits, abrasive grit or shot blast cleaning, and chemical cleaning for sulfide and salt deposits.

Chemical cleaning and steam/air decoking are preferable cleaning methods at an inspection period because they tend to clean the tubes to bare metal. Cleaning to bare metal is imperative when tubes and fittings are subject to spot- or pit-type corrosion. These two cleaning methods can be used to advantage at shutdowns that are only for cleaning, since the coils can be cleaned and returned to service in a short time without unheading.

Chemical cleaning consists of circulating an inhibited acid through the coil until all deposits have been softened and removed. This method is usually followed by water washing to flush all deposits from the coil. Care must be used in chemical cleaning to avoid damage to the tubes.

High-pressure water jet blasting is another option for cleaning tubing with plug-type fittings.

Abrasive blasting (shot blasting or sand jet blasting) with metal shot or an abrasive medium is also a cleaning option for welded coils.

When the tubes are made of austenitic stainless steel, the chloride content of the water used for flushing should be maintained at less than 50 parts per million.

Provision must be made for the safe disposal of the toxic hydrogen sulfide gas generated by the action of acid on sulfide-containing deposits.

Steam/air decoking consists of the use of steam, air, and heat to remove the coke. This method of cleaning should be used only by trained, experienced personnel, since improper procedures or control could result in serious, costly damage to the heater.

Cooling while shutting down and cleaning by chemical or thermal methods may cause leaks in the tube rolls or header plugs of removable headers. These leaks are caused by thermal forces or the removal of coke.

Various types of tube knockers and cutters are available for the mechanical cleaning of tubes. Selection of the type of cleaning head is a matter of preference. The cutting head is usually driven by an air motor. In cold weather, however, steam is often used for motive power to warm the tube and reduce the effect of shock on the tube. When mechanical cleaners are used, care must be exercised to avoid damage to the tubes or fittings.

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Mechanical cleaning cannot be used to clean the U-bends of sectional fittings.

Steam/air decoking will not always remove the coke from a heater fitting. If this is the case, it may be necessary to use mechanical cutters on the U-bends and remove them for cleaning. This is an expensive and destructive method of cleaning.

4.4.3 INTERNAL CLEANING—FIRED BOILERS

Steam-drum internals should be inspected before washing to determine any problems, including poor circulation, poor water quality, and low steam purity.

The inside of shells, drums, and tubes should then be washed down thoroughly to remove mud, loose scale, or similar deposits before they dry and become more difficult to remove. The washing operation should be carried out from above if possible, to carry the material downward to the blowoff or handholes. A hose with sufficient water pressure or hand tools should be used to remove soft scale and sludge. The blowoff line should be disconnected before the washing procedure to keep mud and scale out of the blowdown drum. The tubes of horizontal-return-tube boilers should be washed from below and above. It is especially important to ensure that all tubes and headers are clear of sludge after the wash is completed. Water should be passed down each individual tube and observed to exit from below. Each header should be opened sufficiently to give clear view so that it can be ascertained that all sludge has been removed. Precautions should be taken to ensure that the water does not come into contact with the brickwork of the combustion chamber. If contact cannot be avoided, the brickwork should be dried out carefully when the boiler is fired up.

The use of an inhibited acid solution on the inside of the boiler is becoming a commonly accepted method of cleaning the interior surfaces. After acid cleaning, the interior of the boiler must be neutralized, washed down, and refilled with water. If a nitrogen purge is used after acid cleaning, drums should be checked for oxygen content before entry. Acid cleaning should not be used on superheaters or other equipment which contains pockets that cannot be thoroughly flushed out. Precautions must be taken to make sure that all sludge is removed after an acid wash.

It is normal practice to fill pendent-type superheaters with condensate or demineralized water and to keep the superheater full of this water while the remainder of the boiler is acid cleaned. During chemical cleaning, all phases of the operation should be closely supervised by experienced, responsible individuals. During chemical cleaning, all electric power and other ignition sources in the near boiler must be turned off to prevent explosion of the hydrogen and other hazardous gases that are normally given off during the cleaning.

Another common method of cleaning uses chelates. The chelates are added to the boiler water, and the boiler is fired to create circulation and thereby facilitate cleaning of the internal surfaces.

See the ASME *Boiler and Pressure Vessel Code*, Sections VI and VII, for more information on the care and cleaning of boilers.

SECTION 5—METHODS OF INSPECTION

5.1 Visual Inspection of Heater Coils

5.1.1 GENERAL

When the cleaning operations are completed, the entire heating coil should be given a thorough visual inspection. It is mainly through visual inspection that the effects of deterioration, actual defects, and an indication of potential defects or weaknesses in the tubes, crossovers, fittings, and connections—blowdown, steam, pressure gauge, vents, and thermowell connections—can be found.

5.1.2 EXTERNAL INSPECTION

Tubes should be inspected externally for the following conditions:

a. Sagging or bowing.
b. Bulging.
c. Oxidation or scaling.
d. Cracking or splitting.
e. External corrosion.
f. External deposits.
g. Leaking rolls.

Fittings should be inspected externally for the following conditions:

a. Damage or distortion.
b. Corrosion.

Figures 13 and 14 show examples of the bulging that may occur in tubes, Figure 15 shows an example of scaled tubes, Figure 16 shows an example of an oxidized tube, and Figure 17 shows an example of a split tube. Figure 18 shows examples of the external tube corrosion that may occur during a short shutdown period on a heater that has been fired with a fuel of high sulfur content.

Tubes that have been subjected to excessive temperatures will often sag. In radiant sections, this condition is not considered serious unless it prevents cleaning or causes headers to jam and wedge against other headers or against the sides of the header compartment. In convection sections, sagging

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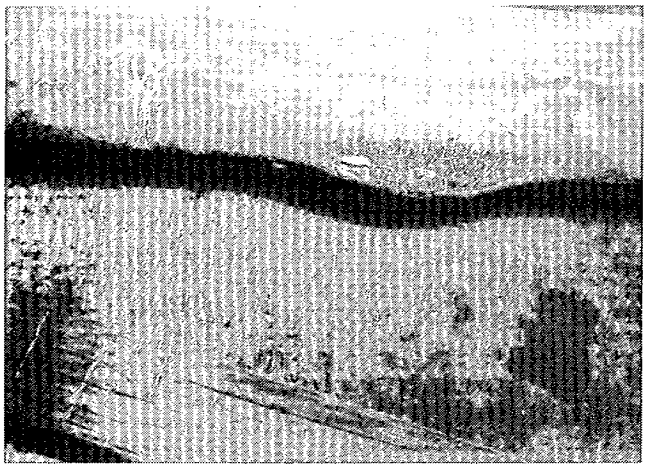


Figure 13—Bulged Tube

of the tubes in upper rows to a point between those in lower rows can prevent the free passage of flue gas around the tubes. This condition, called *nesting*, will cause overheating of adjacent tubes and draft loss. If this condition is found, the offending tubes should be replaced.

A split tube usually results from either localized thinning of the tube wall or a loss of structural strength because of high metal temperature, which may be caused by various factors, including flame impingement and coke buildup.

Bulging is caused by a loss of structural strength, usually as a result of the same conditions cited for a split tube.

Because of the arrangement of the tubes and refractory walls, visual inspection of the external surfaces of the tube is usually restricted to the fireside of the radiant tubes. Special attention should be given to the following locations:

a. The juncture of plain and finned or studded sections.
b. In vertical heaters, the area from the firebox floor to 15 feet above the firebox floor.
c. Entry and exit points through the tube sheets of inlet and outlet tubes.
d. Welds.

When external deterioration, including that due to oxidation, scaling, cracking, and external corrosion, is suspected—especially in the case of convection tubes—representative tubes may be removed from the heater and then cleaned and examined thoroughly. The selection of the tubes to be removed may be guided by the tube locations in the heater, the length of time the tubes have been in service, and the general appearance of the tubes in the area. If the tubes chosen for inspection are found to be defective or unfit for further service, other tubes in the same area and of the same or similar age and general appearance should also be inspected. This should be continued until it is certain that all of the remaining tubes are safe for further service.

When welded tubes are used in a heater, all accessible welds should be thoroughly inspected, regardless of whether they were made by the tubing manufacturer, the pipe fabricator, or plant workers. This inspection should be primarily visual and should be supplemented by magnetic-particle, liquid-penetrant, or radiographic inspection as conditions warrant. The inspection should include both external and accessible internal weld surfaces. The external defects will probably be in the form of cracking, which may be caused by a high metal temperature at the weld.

All tubes rolled into fittings should be examined for leakage in the rolled joint. Leaks in tube rolls and around plugs can often be found by observing the location of coke or oily

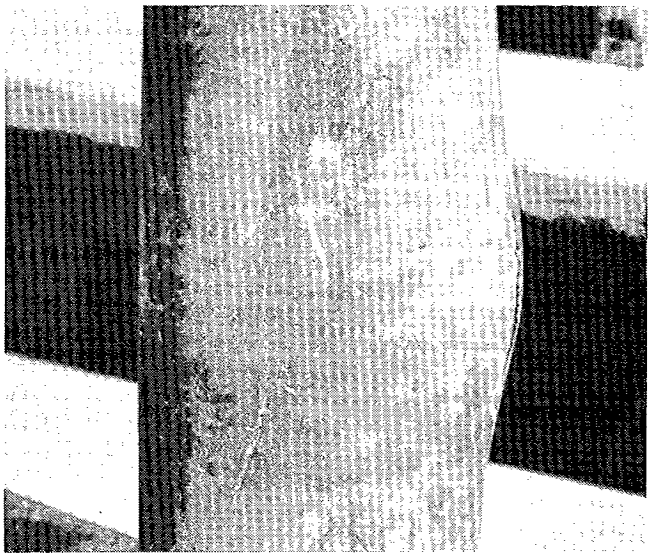


Figure 14—Bulged and Split Tube

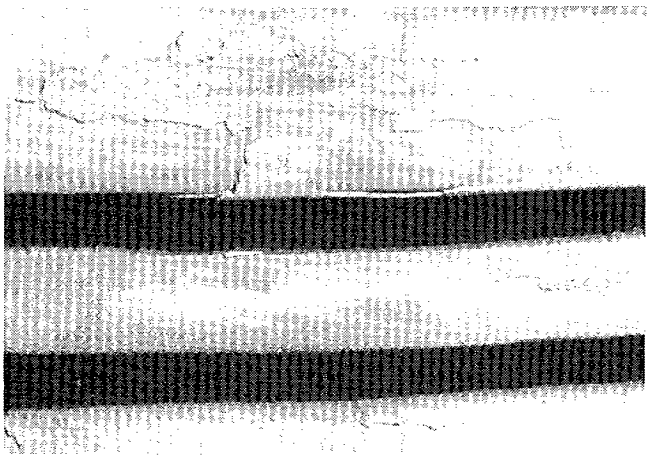


Figure 15—Scaled Tubes

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Figure 16—Oxidized Tube

deposits around headers when the heater is removed from service. An examination should also be made when the coil is under test pressure. The inspection should be visual and should in some cases be supplemented by feeling the tube at the rear face of the fitting for indications of leakage.

Visual inspection can sometimes be facilitated by holding a small mirror between the tube sheet and the fitting to obtain a view of the juncture between the tube and the fitting. Roll leaks will often not become detectable until a coil has been under pressure for 10–15 minutes. Leakage in the tube rolls can be either a nuisance or a serious problem, depending on the operating process and the operating conditions of the heater. Where there is no formation of coke, the leak may be stopped by rerolling the tube. Roll leakage is serious, however, in the case of a heater that is subject to coking and that operates at high pressure-temperature conditions or in poisonous or highly explosive vapor service, including phenol or hydrogen service. Oil leaking between the fitting and the outside surface of the tube can result in the formation of

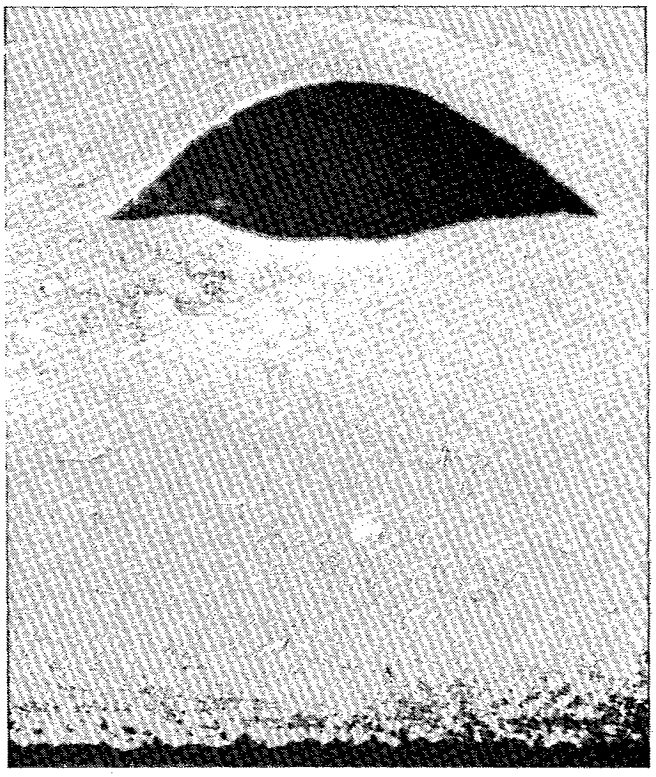


Figure 17—Split Tube

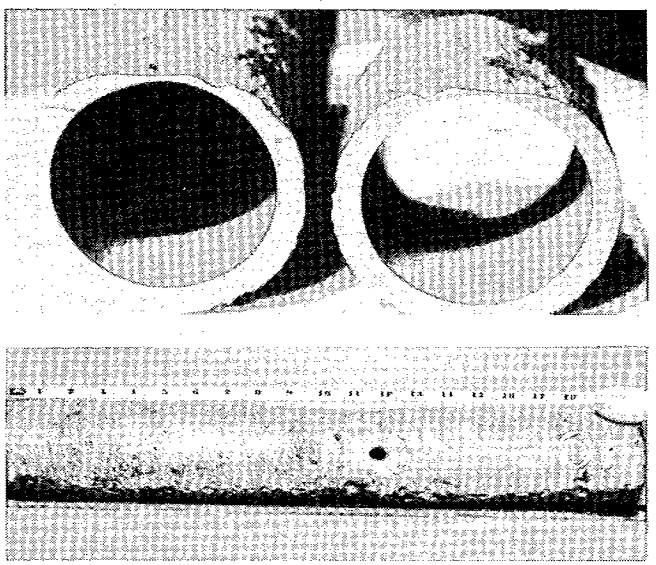


Figure 18—External Corrosion

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coke. This coke formation continues with service, and the force of the coke buildup can be sufficient to cause partial collapse of the tube end and to allow the tube to slip in the fitting. Under these conditions, leakage cannot be corrected by rerolling because the serration in the fitting's tube seat is full of coke, and the mechanical strength of the rolled joint is not improved by the rerolling operation. Figure 19 shows an example of a fitting and a tube that have leaked in the roll.

In the case of fittings, the exterior surfaces of the fitting body and the holding members should be inspected visually. The types of deterioration commonly found on the external surface of fittings are cracking, distortion, and mechanical wear.

Cracking is usually confined to the fitting body or, in the case of welded fittings, to the welded joint. Locations in the fitting body that should be examined for cracking include the area around the plug or U-bend seat, the juncture of an ear or horseshoe holding section and the main body, and the ear or horseshoe section itself. If conditions warrant, a visual inspection of cracks can be supplemented by magnetic-particle or liquid-penetrant examination.

Visual inspection of the ears, the holding members, and the dogs and caps of the holding members is performed primarily to detect distortion and wear, to determine whether there is a proper fit or contact, and to ascertain whether the strength of the fitting has been affected. Figure 20 shows an example of poor fit between the holding section and the cap on a solid fitting.

The threaded portion of the holding screw and the dog or cap should be examined for excessive wear. Distortion that is not apparent to the eye may prevent proper assembly.

The plug or U-bend seat in the fitting should be examined for enlargement, deviations from roundness, change in the width of the seat, and damage to the seating surfaces. The tightness of this joint depends on these four conditions.

For welded fittings, visual inspection is limited to the external surfaces and to the weld attaching the fitting to the tube. The accessible external surfaces of the fitting should be examined closely for any indications of defects, particularly cracks in welds. The inspection of welds should cover a band of 1–2 inches on each side of the weld. Cracks may develop and remain entirely within the weld, or they may start in the weld and run out into the tube or fitting. The inspection of the heat-affected zone and adjacent parent metal is important. It is of paramount importance in the case of alloy welding. The visual inspection of a weld may be supplemented by magnetic-particle, liquid-penetrant, or radiographic inspection.

Crossover sections of tubing used to connect sections of coil may be located outside of the firebox or enclosure but should not be overlooked during inspection of the heater. Movement of the several parts of the coil and changes in temperature can cause stress and fatigue. The surfaces of the tubing, especially bend section surfaces, should be examined for cracks.

5.1.3 INTERNAL INSPECTION

The internal visual inspection of heater tubes is limited to heaters with fittings of the removable U-bend or plug type. On tubes up to about 30 feet in length, it is possible to view



Figure 19—Fitting and Tube That Have Leaked in the Roll

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Figure 20—Spreading and Poor Fit of a Horseshoe Holding Section

the entire interior reasonably well if a light is inserted at the end opposite the one at which the tube is being examined and the examination is made from both ends of the tube.

The inside surface of a tube can be examined with optical instruments. Considerable time is required to inspect each lineal foot of tube. Consequently, optical instruments are generally used for the more thorough inspection of questionable areas revealed by visual inspection.

The internal visual inspection of tubes should be made with the purpose of locating and determining the extent of the following types of deterioration commonly experienced in heater tubes:

a. Selective, spot-type, or pit-type corrosion.
b. Thinning of tube ends.
c. Cutting or other cleaning damage.
d. Loosening of the tube roll and flare.
e. Erosion.

Figure 21 shows examples of the spot- or pit-type corrosion often found in heater tubes. This type of corrosion is one of the most difficult to detect. Visual inspection, internal calipering, and radiography are the only sure means of detection, and even then the internal surfaces of the tubes must be free from coke and any other foreign matter. Mechanical cleaning will not always reveal spot- or pit-type corrosion. If this type of corrosion is apparent or suspected, the inside surfaces of the tube at the tube ends must be cleaned using an acetylene torch to burn coke or other foreign matter out of the pits.

Thinning at the ends of rolled-in tubes is usually caused by erosion or turbulence that results from change in the direction of flow. This type of thinning may also result from frequent rerolling of tubes to stop leakage.

Figure 22 shows an example of a tube damaged by a cleaning head. In some cases the outside diameter of the tube may be increased and will have the same general appearance as a tube with a slight bulge.

Figure 23 shows an example of eccentric corrosion of a tube. The loss of wall thickness is not uniform around the circumference. In this type of deterioration the most thinning usually occurs on the fireside of the tube. This type of corrosion is generally accelerated on the fireside because of the high metal temperature there. Eccentric corrosion may also be caused by external scaling. It is often difficult to determine whether tubes have become eccentric as a result of service, since the condition is not readily detectable by visual inspection of the tube ends. An indication of eccentric corrosion can sometimes be found by measuring several diameters at one location. It is difficult to detect by hammer testing. A reliable means of detection is to measure thickness with ultrasonic or radiation-type instruments, but these tools can only be used on accessible tubes, usually the radiant tubes. Although this type of corrosion is more common on radiant tubes, it has occurred on convection tubes, usually on those adjacent to the refractory.

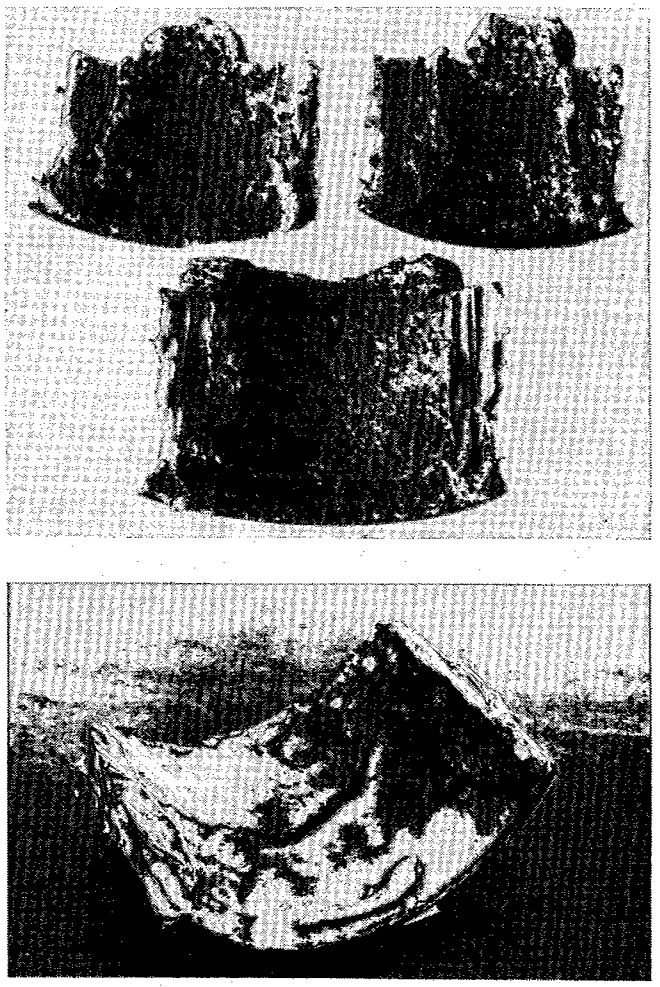


Figure 21—Spot- and Pit-Type Corrosion

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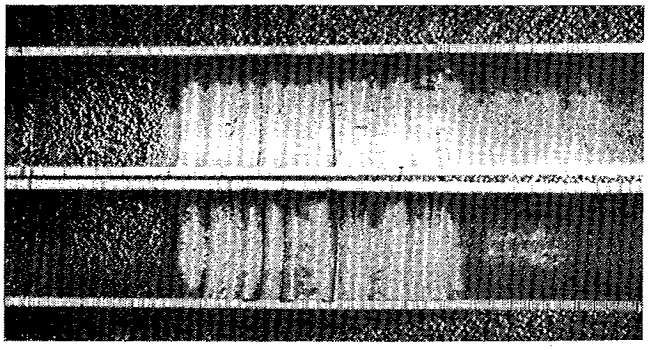


Figure 22—Tube Damage Caused by Mechanical Cleaning Equipment

In some heaters, part of the tubes may not be accessible for an internal visual inspection. As a substitute for internal visual inspection, some companies make a practice of thoroughly inspecting all tubes that are condemned and removed from a heater, regardless of the reason for the tubes' removal. This inspection is made by cutting a tube into short sections of 2–3 feet so that the inside surface can be examined. Measurements for metal-wall thicknesses are made at the ends of each section. In some cases the sections are split longitudinally, thus exposing the entire inside surface for examination. The ends of the tube rolled into the fitting should be removed for examination. They may then be inspected to determine the general condition and effectiveness of the rolled joint.

In some cases, a rolled-in tube may also be welded to the fitting. There are two basic reasons for welding a tube to the fitting: (a) to stop leakage by means of a seal weld and (b) to improve the efficiency of the rolled joint by means of a strength weld. The use of a strength weld warrants careful consideration and justification. Any welding between the tube and the fitting, regardless of its basic purpose, should be examined carefully.

The types of defects that are commonly found are cracking, slag, and porosity in the weld.

In the case of rolled-on fittings, the internal surface should be inspected visually for signs of deterioration and to ascertain the fittings' general physical condition. With sectional, streamlined fittings, the housing section (the part the tube is rolled in) should be examined for undercutting, the width and condition of the U-bend seats, and excessive erosion and thinning of the housing in the annular space (the section of the housing between the end of the tube and the inside edge of the U-bend seat). The inside surfaces of the U bend should be examined for thinning and to ascertain their general condition.

With solid fittings, the body section should be examined for undercutting, the width and condition of the plug seat, and erosion and thinning of the barrel section of the body (the cylindrical section with the plug seat at one end and the tube seat at the other end) and the cross port (the connecting section between the two barrel sections).

Figure 24 is a sectional view of a streamlined fitting. It shows the severe corrosion-erosion that can occur in the annular space and at the inside edge of the U-bend seat. The seating face on U bends and plugs should be examined for corrosion, and the width of the seat should be checked against the width of the seat in the housing or body sections. If there is not a tight fit between the U bends and the housing for the entire width of the seating surface or if the width of the seating surface is longer on one member, erosion of the members will be severe. This same condition should be checked on solid fittings at the closure area between the fitting body and the plug. Fittings should be examined to determine the fit and depth of seating between the U bend or plug and the main body of the fitting. If the fitting seat has become enlarged through service, the U bend or plug can protrude so deeply into the fitting that it is not possible to head up and get a tight joint when the fitting is under pressure. In the case of a sectional fitting, the end of the U bend will contact the end of the tube or the tube stop, depending on the type of tube seat used. In the case of a solid fitting, the ears on the plug will contact the outside face of the fitting. Figure 25 shows an example of the type of corrosion experienced in U bends.

5.2 Visual Inspection of Fired Boilers

5.2.1 PRELIMINARY INSPECTION

It is good practice to make a preliminary inspection of the inside of all equipment to the extent practicable before the boiler is cleaned. The location, amount, physical appearance, and analysis of mud, sludge, or scale deposited on the inside

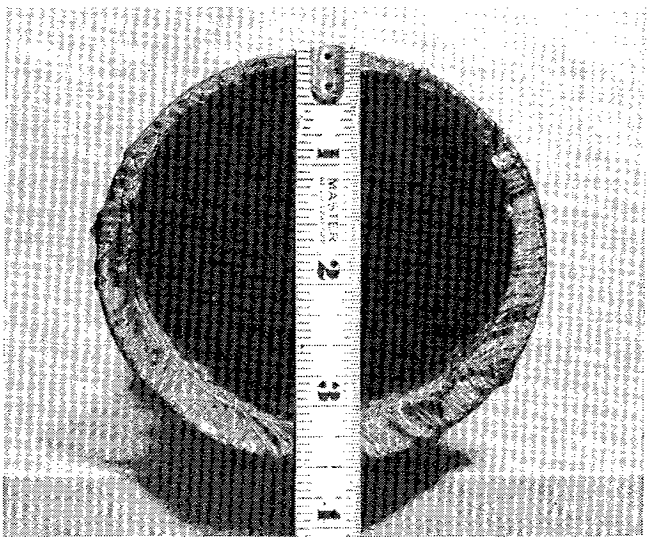


Figure 23—Eccentric Corrosion of a Tube

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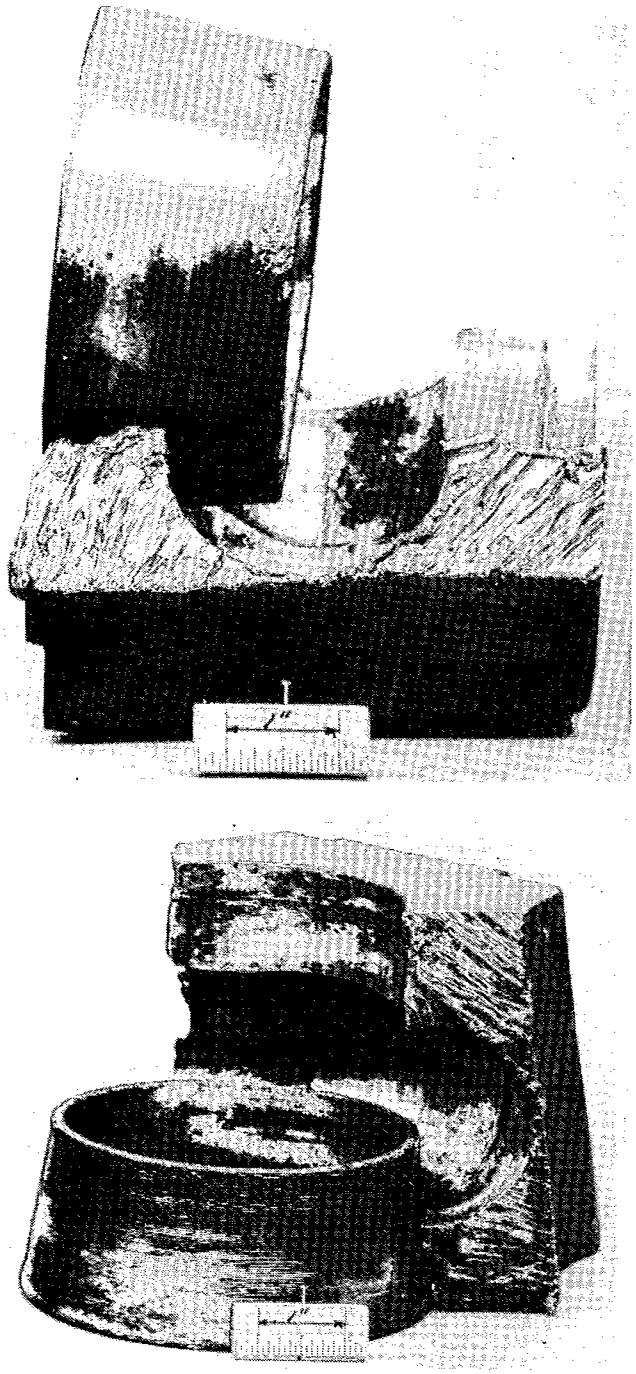


Figure 24—Corrosion/Erosion of the Annular Space in a Streamlined Fitting

of shells and drums will provide information about the effectiveness of the feedwater treatment, blowdown operation, and methods of cleaning needed. The preliminary inspection may also be helpful in determining which parts of shells or drums require the closest inspection. Heavy scale found either in drums or in or on tubes should always be a signal to inspect the scaled area closely for metal overheating. Flow marks in fly ash or soot deposited on the baffling may be of great help in locating gas leaks in the baffling. Any conditions which indicate that close inspection is required after cleaning should be noted.

After the preliminary internal inspection and general cleanout, the detailed inspection may proceed. If welded seams are heavily coated, they may have to be sandblasted or scraped and wire-brushed before a visual examination is possible. Ordinarily, it is not necessary to remove insulation material, masonry, or fixed parts of the boiler, unless defects or deterioration peculiar to certain types of boilers are suspected. Where moisture or vapor shows through the covering, the covering should be removed and a complete investigation made.

5.2.2 PIPING, PIPE JOINTS, AND REFRACTORY LINING

A visual inspection should be made for evidence of leakage in pipe and threaded or flanged pipe joints. Water leaks may be detected by the presence of moisture or deposits at the point of leakage and steam leaks by the appearance of the adjacent metal.

Leaks may sometimes be a result of strains caused by deformation or misalignment of the piping system. Deformations may be caused by lack of provision for expansion or by

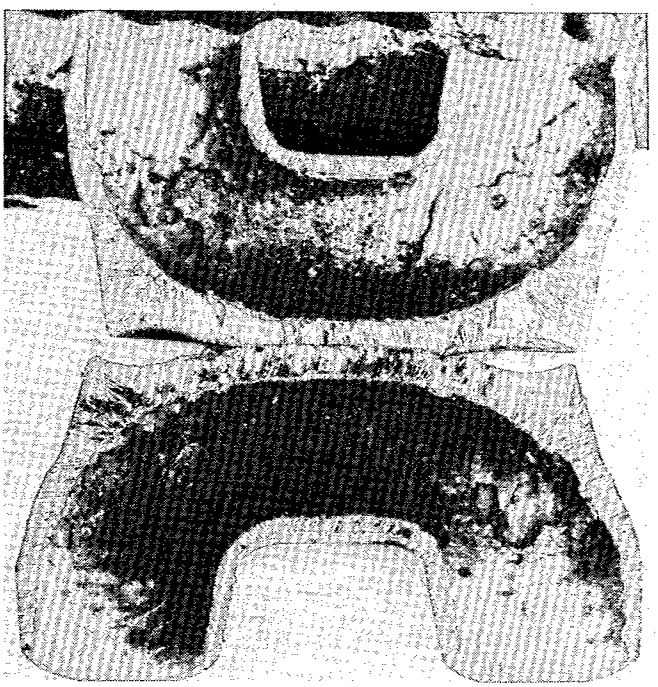


Figure 25—Corrosion of U Bends

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improper supports. If not eliminated, pronounced deformation may place strains of sufficient magnitude to cause failure in small connections. It is important that a careful inspection be made to determine if such defects are present.

When flanged connections are opened, gaskets and gasket seats should be inspected carefully. Gaskets may be damaged by leakage or by improper centering of the gasket when the joint is made up. Gasket seats may be scored by a steam leak at the joint, improper handling, or careless use of tools. Seating surfaces should be inspected for tool marks, other mechanical abuses, and evidence of the type of erosion commonly called *steam cutting* or *wire drawing*. Mechanical damage may lead to erosion if not corrected. Either a scored seat must be machined to provide a proper gasket face or the flange must be replaced; otherwise, leaks will recur. Before joints are remade, ring gaskets should be examined to determine their fitness for reuse. Other types of gaskets should be replaced with new ones.

The condition of refractory linings in the combustion chamber, stacks, flue-gas ducts, observation and access doors, and around burner ports should be inspected. Special attention should be given to the lining sections intended to protect pressure parts and supports from overheating. If any of the refractory in the combustion chamber has fallen out, the supporting steel will be exposed to excessive temperatures that will damage the steel. Linings in stacks and ducts may also have areas where the refractory has fallen out. When this occurs, the outer structure is exposed to temperatures that are greater, in most cases, than the material is capable of withstanding. Outer structures composed of brick will develop cracks; outer structures composed of steel will buckle. Eventually, failures will occur unless corrective measures are taken to replace the refractory. Entrance of air into a boiler furnace or stack, other than through the burners or related openings, may cause inefficient and potentially dangerous boiler operating conditions.

A visual survey of the heater or furnace should be made for air leakage into a balanced draft unit and for leakage out of a positive pressure unit. Cracks and loose access and fire doors, peepholes, and joints permit air leakage. An artificial smoke source—titanium tetrachloride, hydrated zinc chloride, or another source of smoke—placed close to the cracks may be useful for the inspection. Use of smoke for the inspection should be done with due consideration of the hazards associated with the materials and the appropriate personnel safety equipment. See following Note. Leakage into the furnace or heater, when such leaks are adjacent to the structural steel supports, may result in temperature gradients of sufficient intensity to cause failure of the supports. This is particularly likely to occur in areas where combustion is not complete and the concentration of carbon monoxide is high.

Note: The Material Safety Data Sheet for the type of smoke used should be consulted.

5.2.3 INTERNAL INSPECTION OF BOILER COMPONENTS

5.2.3.1 General

All manhole covers and a sufficient number of handhole plates should be removed for inspection. During inspection, all parts should be observed with the thought of proper operation in mind.

5.2.3.2 Drums, Drum Connections, and Internal Parts

All internal surfaces and the connections to all outside attachments, including water-column connections and safety-valve nozzles, should be examined for deformation, corrosion, pitting, grooving, scale deposits, and sludge accumulation. Special attention should be paid to all seams, whether welded or riveted, and to the areas adjacent to them. Welded seams and connections should be examined for cracks. The welded seams in deaerators should be cleaned, prepared, and inspected by wet fluorescent magnetic-particle examination.

Riveted joints should be checked for loose or broken rivets, cracking, or other evidence of distress. Rivets should be hammer-tested for soundness. If there is any evidence of leakage or other distress in lap joints, it should be investigated thoroughly, and if necessary, rivets should be removed or the plate should be slotted to determine whether cracks exist in the seam. The top external surface of mud drums should be cleaned of all deposits, and the surface should be examined for corrosion. Corrosion along or immediately adjacent to a seam may be more serious than a similar amount of corrosion in the solid plate away from the seams. Grooving and cracks along longitudinal seams are especially significant, as they are likely to occur when the material is highly stressed. Severe corrosion is likely to occur at points where the circulation of water is poor. Such points should be examined carefully, and the minimum remaining thickness should be determined by ultrasonic technique and then recorded. Inspection of the steam drum should include observations of the normal water level. Any bulges or uneven areas that would indicate excessive heat input from leaking fireside bafflers should be noted. Evidence of poor circulation may be indicated by waterline gouging along the top half of the top one or two rows of downcomers. This is sometimes accompanied by flash marks on the drum surface at the tube openings. If a sample of the boiler drum is needed for chemical analysis or microscopic examination, a section may be trepanned from the wall. The resulting cavity must be repaired by a suitable method such as welding. Normally, an ultrasonic technique is used to measure wall thickness. Occasionally, a hammer test may be used to locate thin areas in the drum plate. These areas should then be measured by ultrasonic technique.

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When a more thorough examination for cracks and other defects in plate and weld metal is desired than can be obtained by a visual inspection, a radiographic, magnetic-particle, ultrasonic, or dye-penetrant test may be used as follows:

a. The radiographic test is used to examine suspected areas for defects and cracks below, on, or near the metal surface.
b. The dry powder magnetic-particle test is used to determine cracks on or near the surface.
c. The wet fluorescent magnetic-particle test uses a black light for finding discontinuities and can locate fine surface cracks that may occur in deaerator welds.
d. The ultrasonic test is used to indicate discontinuities in the metal at any depth.
e. The dye-penetrant test is used to locate surface cracks in large or small areas.

Drum internals and connections to the drum should be inspected when the drum is inspected. Welds or rivets attaching internals or connections to the drums should be inspected in the same manner as welds or rivets in the drum proper.

Safety-valve nozzles and gauge-glass connections, especially the lower connections, should be examined for accumulations of sludge or foreign material. A flashlight should be used to visually inspect the nozzle or connection. If the inside cannot be observed directly, a small hand mirror may be used for indirect observation. Special forms of illuminating equipment, mirrors, and magnifying devices are very useful for this type of inspection. When the boiler contains more than one drum, usually only one of the drums will have safety valves on it.

Any manhole davits should be tested for freedom of movement and for excessive deformation. Manhole and handhole cover plates and nozzle seats should be examined for scoring in the manner described in preceding text for pipe flanges. Cover plates should be inspected for cracks.

Drum internals, including internal feed header, distribution piping, steam separators, dry pipes, blowdown piping, deflector plates, and baffle plates, should be inspected and hammer-tested for tightness, soundness, and structural stability. The vigorous turbulence of the steam and water mixture present in the drum may vibrate such parts loose from their fasteners, attachments, or settings. When these parts are welded in place, it is not uncommon for the welds to crack from vibration. Steam separators and baffles should be carefully inspected for tightness, corrosion, and deterioration, and associated welds should be checked for cracks. Any bypassing of the steam separator will permit carryover into the superheater, causing salt deposition, resultant overheating, and possible tube failure. Steam separators should be free from deposits that might impair their operation. To ensure that the proper points are observed, inspection personnel should become familiar with the operation of the type of steam separator used in the boiler. Some boilers do not have steam separators and depend entirely on dry pipes for water separation.

The holes in dry pipes should be free from any deposits that might restrict flow. Since dry pipe holes are in the top of the pipe near the top of the drum, it may be necessary to inspect the holes indirectly with a hand mirror. Any drain holes in the pipe should also be inspected for freedom from deposits and scale. Not all drums contain dry pipes.

Tubes, downcomers, and risers should be inspected for scale or deposit buildup, erosion, and corrosion. Tubes should be checked for any cutting. Figure 26 is a photograph of the interior surface of a tube that has been damaged by operating a tube cleaner too long in one place.

Ultrasonic testing and hammer sounding are good methods of checking for tube wall loss caused by corrosion. Tube ligaments should be examined for cracks. If tubes are covered by baffle or deflector plates, a few of these plates should be removed to permit a spot check of the condition of the tubes behind them.

The methods described in API Recommended Practice 572 are applicable to all drums forming any part of a steam boiler.

5.2.3.3 Water Headers

Each handhole and handhole plate seat should be examined for erosion, steam cutting, tool marks, and other abuses that might permit leakage. If the plate has leaked previously, it should be checked for trueness and possible deformation. Seating surfaces and faces of handholes should be examined for cracks. It may be necessary to use a hand mirror to inspect the handhole seats.

The inside surface of the headers should be inspected for corrosion and erosion. The location and amount of scale

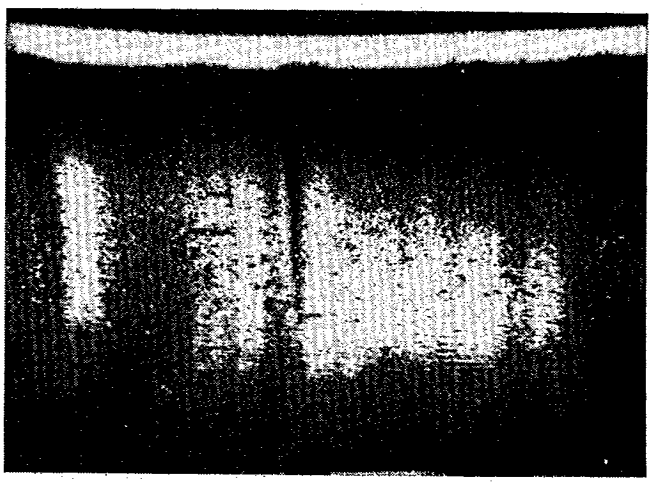


Figure 26—Interior Surface of a Tube Damaged by Operating a Tube Cleaner Too Long in One Place

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buildup should be noted, and the tube ends should be checked for pits, scale, cutting or other damage from tube cleaners, and deposit buildup. If there is considerable scale or deposit buildup, the flow may be restricted to the point that tubes become overheated because of insufficient circulation. Deposits and scale should be removed with a scraper and the depth of coating determined. Lower waterwall headers are particularly susceptible to heavy deposit buildup.

Downcomers and risers should also be inspected for this type of deposit. Thickness readings of headers should be obtained periodically by ultrasonic technique. The headers should be calipered whenever tubes are removed.

External surfaces of headers should be examined either directly or indirectly with mirrors, and particular attention should be paid to the points where tubes enter the header for indications of leakage from the tube roll. The header surfaces adjacent to tube rolls and handholes should be inspected for cracks. If external inspection of headers reveals pitting, thickness measurements should be made using ultrasonic techniques.

5.2.3.4 Superheater Header

Except as indicated in text that follows, inspections of superheater headers should be conducted in a manner similar to that for inspections of waterwall headers.

Usually, superheater handholes are not opened at every boiler shutdown or cleanout unless tubes are to be replaced or other repairs are to be made. For a spot check, however, a few of the handholes should be removed at every shutdown.

Since only dry steam passes through the superheater, there should be few or no deposits present in the headers or tubes. If deposits or scale are present in any degree, immediate steps should be taken to determine why they are present. In addition, the extent of the deposits or scale should be investigated. Superheater tubes with a moderate deposit of scale will rupture readily from effects of overheating. Indications of scale or deposits should lead to an investigation of the steam separators, dry box, operating drum level and fluctuations, blowdown rates, and water quality.

5.2.3.5 Deaerators and Deaerator Storage Vessels

Deaerators and deaerator storage vessels should be inspected in a manner similar to that for the investigation of any pressure vessel (see API Recommended Practice 572).

Particular attention should be given to weld inspection. The internal longitudinal and circumferential welds and their heat-affected zones should be checked carefully for cracks running in both longitudinal and transverse directions. The recommended inspection method is wet fluorescent magnetic-particle examination.

Proper cleaning of the welds for inspection is required. Abrasive blasting or grinding to a smooth finish is recommended to properly clean the welds for inspection. If grinding is used, care must be exercised to ensure that no defects are hidden.

5.2.4 EXTERNAL INSPECTION OF BOILER FIRESIDE COMPONENTS

5.2.4.1 General

The firebox may be entered through an access door or by removing a burner. A person should be stationed outside the firebox continuously so that he can always see the workers inspecting the inside of the firebox.

5.2.4.2 Refractory Linings

Refractory linings should be inspected for cracks, erosion, excessive fluxing (melting of the refractory), bulging, and fallout. Cracks in the refractory are common and are to be expected. Only the degree of cracking is important. If the refractory is severely cracked, repairs should be made. No rules or limits can be established indicating what can or cannot be tolerated. Decisions have to be made based on good judgment and good practice.

The presence and extent of refractory erosion or fluxing should be determined. Metal parts and insulation behind the refractory will become overheated and damaged if these conditions are permitted to remain unchecked. When excessive erosion or fluxing occurs in the lower section of a wall, the upper sections may be undermined to the point that they will fall out because of insufficient support.

Erosion is caused by flame impingement, high ash velocities, and inferior materials. Erosion may occur around burner throats, furnace sidewalls, and furnace back walls. In boilers with waterwalls, erosion tends to occur in the refractory material between the tubes, especially on back walls opposite burners.

Fluxing is caused by inferior or improper materials, ash containing metal oxides, or flame impingement. Fluxing may occur at almost any point, but locations in the direct path of the hot gases would be most susceptible to fluxing.

The depth of erosion or fluxing and the remaining thickness of the refractory should be measured. The depth of local erosion or fluxing may be measured with a straight edge and rule. In areas around burner throats, the extent of erosion or fluxing may be difficult to determine because of the circular or conical shape. Photographs or blueprints of the original installation are helpful references in establishing the extent of erosion in these areas. The thickness of the remaining refractory may be measured by drilling or cutting out a small piece in the suspected area.

Refractory that has fallen out or bulged to the point that it is in danger of falling out should be replaced. The area replaced should be in the form of a square or rectangle. The edges should be cut straight in and not tapered. An area of

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about 12 square inches (30.5 square centimeters) should be the minimum area cut out and replaced. Bulging and fallout may be due to settlement of the anchor bolts, anchor brackets, or castings or of the furnace-setting supports themselves. When bulging or fallout is encountered, the cause should be ascertained so that corrective measures may be taken to prevent a recurrence. Bulging or fallout in waterwalls may be due to failure of the tubes to transfer the severe heat. In rare cases this may be caused by too large a tube spacing, but it is generally caused by blocked or clogged tubes.

5.2.4.3 Tubes

All tubes should be inspected for signs of overheating, corrosion, and erosion. Usually, overheating is caused by deposits or excessive scale on the waterside of the tube. Waterwall tubes and generating tubes nearest the furnace are particularly susceptible to overheating and should be closely examined for bulging, blistering, quench cracking, sagging, and bowing.

Boiler tubes should be inspected at the steam-drum connection for gouging and caustic corrosion due to steam blanketing. Roof tubes are generally designed for heat pickup on one side only. Therefore, a sagging roof tube due to burned-out hangers is especially susceptible to overheating. These tubes should be straightened, and the hangers should be replaced.

Inspection for blisters and local bulging is easily accomplished by shining a flashlight parallel to the length of the tube so that bulges, blisters, and other deformities cast shadows. Cleaning of a slagged tube may be necessary to find minor blisters. The tube's outside diameter should be measured across the blister or bulge. If the reading is equal to the tube outside diameter plus 5 percent or more, then the distorted area should be replaced or properly repaired.

Waterside corrosion, generally caused by faulty water treatment, can usually be detected by ultrasonic or hammer testing as discussed in preceding text. A few selected tubes should be ultrasonically measured for minimum thickness. Measurement can also be made from inside the steam drum for a distance of 8–10 inches into the tubes. The locations measured and thicknesses found should be recorded to establish a tube corrosion rate. Fireside corrosion is generally caused by moisture that accumulates in fly-ash deposits. Although fireside corrosion may occur anywhere in the tube nest, it usually occurs where the tubes enter the lower drums or headers.

Moisture causing fireside corrosion can come from leaks in tubes, drums, headers, faulty steam soot-blower shutoff valves, from rain water through stacks and roofs, and from condensation from the atmosphere during downtime.

Because boiler tubes usually are not very thick, corrosion can be serious. The tubes should be examined for corrosion. A scraper should be used when examining for external corrosion.

When a tube rupture occurs, the tube should be visually inspected. Its appearance may indicate the cause of failure. If the cause is not evident, samples of the tube in the original condition, with deposits and scale intact, should be taken and analyzed chemically and microscopically. The tube sample should be cut at least 1 foot on either side of the failure.

The inside of bent tubes and of straight tubes, as far as it is accessible, should be examined with strong illumination. Straight tubes should be examined by illuminating the end away from the observer.

Internal cleanliness is required to achieve a satisfactory inspection of tubes. In case of doubt concerning the cleanliness of tubes, a turbine-type cleaner should be put through representative tubes, and the loosened deposits should be trapped at the discharge ends. The weight of trapped deposit and the internal surface area will indicate the average thickness of the deposit removed. Fiber optics or borescopes—borescopes are of limited use on bent tubes but are satisfactory for viewing straight tubes—may also be used to inspect tube internals. Tube ends should be checked for proper projection and flaring. Calipers, micrometers, and ultrasonic instruments can be used to measure tube diameters, dimensions of bulges on tubes, depth of corrosion pits, and tube-wall thickness. These measurements are of great value in determining the effects of corrosion and erosion and in estimating the future lives of the parts measured. Erosion of exterior surfaces is caused by the impingement of fly ash or raw fuel solids at excessive velocity or by soot blowers. Fly-ash tube erosion can be arrested by installing shields or by reducing the gas velocity. If erosion is due to soot-blower medium impingement, the soot blowers should be checked for alignment, warpage, and operating wear. Wastage of exterior tube surfaces can be caused by flame impingement, which should be corrected by adjustments to the firing equipment.

Some types of waterwalls have tubes widely spaced and the area between the tubes covered by steel fins attached to the tubes. The fins may become overheated and burn or crack. The fins should be inspected for cracks that may extend into the tubes. The tubes should be inspected for signs of leakage that may result from the cracks.

Waterwall tubes should also be checked for alignment. All gas passages should be inspected for slagging or bridging from fly ash or slag buildup. The first gas pass is particularly susceptible to this condition.

See the ASME *Boiler and Pressure Vessel Code*, Sections VI and VII, for more information on inspection of boilers.

5.3 Determination of Wall Thickness

The determination of the wall thickness of the tubes and fittings in a heater is an essential feature of inspection. These wall thicknesses provide a record of the amount of thickness lost, the rate of loss, the remaining corrosion al-

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lowance, the adequacy of the remaining thickness for the operating conditions, and the expected rate of loss during the next operating period. The two basic types of methods used to determine the wall thicknesses of piping and tubes are the following:

a. Destructive methods. One destructive method is removal of any tubes that are deep in convection banks and inaccessible for measurement of the tube walls using calipers.
b. Nondestructive methods. These include the following:
   1. Measurement of the inside and outside diameters of the tube.
   2. Measurement by means of ultrasonic instruments.
   3. Measurement by means of radiation-type instruments or radiography.

A wall thickness value obtained by measuring the inside and outside tube diameters is called an *indicated metal thickness*. The indicated metal thickness is obtained by measuring the inside and outside diameters of the tube at several locations—in most cases the outside diameter is not actually measured but is taken as the nominal outside tube diameter—subtracting the maximum inside diameter from the corresponding outside diameter, and dividing the difference by 2. If the tube happens to be eccentric, the indicated metal thickness value can be very much in error. A tube operating in an area of high creep stress will increase in diameter after prolonged service. An increase in outside diameter can cause an error unless outside diameters are directly measured.

There are many types of calipers for measuring the inside diameters of heater tubes, including the simple 36-inch (91.44-centimeter) mechanical scissor and the 2-point pistol type, the cone or piston type, and the 4–12-point electric type. A caliper equipped to measure several diameters around the circumference of a tube is more likely than others to find the actual maximum inside diameter.

Bench marks on heater tubes are sometimes used to determine the amount of external scaling. Two holes 7/16 inch (1.11 centimeters) in diameter spaced approximately 4 inches apart are drilled in the tube on a line parallel to the axis. The depth of the holes should not exceed the established minimum allowable thickness of the tube. These two holes are filled with either Inconel or 25-20 stainless steel weld metal, which is ground flush with the surface of the tube. As the tube scales externally, the amount of loss may be measured by placing a straight edge between the two buttons of alloy weld metal and using a rule to measure from the straight edge to the tube wall. Bench marks are only installed on tubes that have experienced an excessive amount of scaling and where there is a possibility of flame impingement on the tubes. Each of the three methods of determining wall thickness—measuring the inside and outside diameters of tubes, measuring by means of ultrasonic instruments, and measuring by means of radiation-type instruments or radiography—can be used to check the thickness of heater tubes.

Calipering inside diameters is usually restricted to tubes with removable U-bend or plug-type fittings. It is general practice to caliper the inside diameter of a tube at two locations: in the roll and in back of the roll. Since an increase in internal diameter may not be uniform throughout the length of the tube because of erosion, erratic corrosion, bulging, or mechanical damage while cleaning, it is advisable to take several measurements to determine the worst section of each tube. On heaters where the pattern of corrosion is uniform and well established and mechanical damage is known not to exist, measurements for approximately 36 inches (91.44 centimeters) into the tube may suffice.

The roll section of a tube in service should be calipered to locate the maximum inside diameter at any point between the back edge of the tube flare, or the end of the tube if there is no tube flare, and the rear face of the fitting or edge of shoulder left in the tube by the rolling tool.

As a result of varying shapes, limited working space, obstructions, and the like, it is very difficult to examine the various sections of heater fittings to determine accurately the point of minimum wall thickness. In the case of the rolled-on type of fitting, the easiest and most commonly used method is to use a C-type direct-reading caliper. There are many C-type calipers to choose from.

Ultrasonic methods for obtaining tube-wall thickness are probably the most widely used methods. For most corrosion inspection, straight-beam ultrasonic techniques are used. The sound is introduced perpendicular to the entrance surface and reflects from the back surface, which is usually more or less parallel to the entrance surface. On thick-wall materials, single-crystal transducers are usually preferable. On thinner materials or under other special conditions, dual probes are more desirable, since they produce cleaner, more usable signals, especially from rough or nonparallel surfaces. Dual probes provide the instrument with the capability to increase the gain and thereby improve the probability of detecting small, pitting-type reflectors. Dual probes are usually used for heater or boiler tube measurements.

Radiography of tubing can show variation in thickness of a minimum of 2 percent of total thickness. Thickness is determined by directing the rays tangentially to the tube wall and recording the radiation on a film behind the tube. By comparing with some geometric standard projected on the film, the wall thickness can be determined. Radiographic techniques are particularly useful where the coil has welded return bends. A benefit of radiographic techniques is that they frequently reveal internal deposits in tubing.

5.4 Other Types of Tests and Examinations

5.4.1 METALLURGICAL TESTS

It has already been stated that certain types of deterioration experienced in heater tubes result from some change in

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metallurgical structure. The more common types of deterioration are carburization, decarburization, the initial stages of stress cracking, fatigue cracking, and some forms of hydrogen attack. More information about conditions that cause deterioration or failure can be found in API Recommended Practice 571.

It is possible to detect most of these types of deterioration in the field by visual inspection, nondestructive testing, in-situ metallography, or replication. Carburization and decarburization can only be determined accurately by a chemical or physical test. Most of the testing must be done by specially trained personnel. Damage that results from some metallurgical changes can be determined by ultrasonic, magnetic-particle, and liquid-penetrant testing.

Examination for the forms of deterioration mentioned in the preceding text can be performed on specimens taken from tubes that have been condemned and removed from a heater. In some cases, conditions may warrant the removal of representative tubes from the heater solely to make these examinations. If equipment is not available locally to make the required types of examinations, there are commercial laboratories that specialize in this type of work.

5.4.2 MAGNETIC TEST FOR CARBURIZATION OF AUSTENITIC TUBES IN PYROLYSIS FURNACES

Austenitic tubes are essentially nonmagnetic. Carburized areas of the tubes become magnetic, and if these areas are large, they can be detected with a magnet. A magnet on a string dropped down a tube will indicate areas that are magnetic but will not indicate the depth of carburization. Some instruments and field services can relate the degree of magnetism to the depth of carburization. Most of the instruments are proprietary, and the field services are limited.

A rule of thumb states that up to 50-percent carburization can be tolerated on stream before loss of strength materially affects tube life. Although this rule of thumb indicates that a tube with 50-percent carburization should be replaced, it does not mean that less than 50-percent carburization will allow the tube to remain in service until the next shutdown. Factors including the rate of carburization, the expected service time until the next shutdown, the amount of excess metal, and changes in pressure and temperature must be taken into account.

5.4.3 ULTRASONIC INSPECTION FOR STRESS RUPTURE CRACKING

Stress rupture cracking of cast tubing used in steam/methane-reforming and pyrolysis furnaces usually starts at the midwall of the furnace tube and is normally longitudinal, resulting from hoop stresses in the tube.

Ultrasonic equipment that implements through transmission (pitch catch) has been used to inspect tubes. With this method, a grading of percent transmission is made to evaluate the degree of fissuring, which impedes transmission of the ultrasound. Since tubes vary in the amount of equiaxed and columnar grains, the standard used should reflect the tubes being inspected. Without an adequate standard, the judgment of percent transmission may be in error.

Evaluations of tubes have indicated that the initiation of internal fissuring will eventually cause the tube to fail, but probably not for 30,000–50,000 hours. Major fissuring, which is easily detected, indicates that failure may occur in up to 10,000 hours. Since such a wide range of tube life is available for evaluation, a risk analysis should be made. Tubes that are expected to fail in less than 1 year should be replaced. Tubes that may be good for several years may be allowed to remain in service until the next scheduled shutdown, when they can be reinspected or replaced. Replacement tubes can be ordered and would be on hand when needed. All these evaluations must be based on the assumptions that the original design and casting quality are adequate and that operation, especially with respect to tube metal temperature, is within the design limits.

Furnaces with external pigtails have been operated to tube rupture. In such cases, pigtail nipping has been used to crimp the inlet and outlet pigtails to cut off the inlet and outlet gases. Designs for pigtail nippers are available but should be checked to ensure that the hydraulic-system pressure is enough to cut off all flow (usually over 5000 pounds per square inch gauge), that stops are on the anvils to prevent the pigtail from being cut off (the design should be based on wall thickness), and that some locking device is available to keep the crimp closed when the pigtail nipper pressure is released for removal of the hydraulic cylinders.

5.4.4 RADIOGRAPHIC INSPECTION OF REFORMING TUBES

Radiographic methods have been used to inspect reforming tubes. However, tight cracks cannot readily be seen unless they are normal to the film. When catalyst is in the tubes, the tight cracks will be harder to find because of the varied film densities and the catalyst edges that are present. It is desirable to remove the catalyst from the tubes, but this is not normally practical or economical when the catalyst is not scheduled for replacement.

Radiographs can show cracks regardless of whether there is catalyst in the tubes. However, radiography may not be as sensitive to initial fissuring and tight cracks as is ultrasonic inspection. If radiographs do show cracks, the cracks can be judged on the basis of how many there are and how wide they appear to be on the radiograph. Normally, dark, wide cracks on a radiograph indicate that the cracks are open to the inside diameter of the tube and that the tube should be replaced.

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5.4.5 HAMMER TESTING

A hammer test is an accepted method of exploring the surface of metal objects to locate areas of reduced wall thickness. When a hammer test is made, the variations in metal-wall thickness are indicated by the feel and rebound of the hammer and by the sound produced.

The value of the hammer test depends on the experience of the person who performs the test. More skill is required to hammer test tubing than to test plate. This is because the resilience varies with the size of the pipe or tube and with a change in the tube material. The feel and rebound of the hammer are an indicated measure of the rigidity of the tube or pipe. In the case of exceptionally thin areas, the surface of the tube or pipe may be dented by the hammer. The fireside of the tube should be explored carefully for signs of thinning.

Hammer testing is a good way to determine whether the scale on the outside surface of a tube is an oxide due to overheating or a product of fuel combustion. Although combustion deposits may vary in texture depending on the fuel used, the scale that results from oxidation is generally harder, requires a stronger blow to be knocked loose from the tube, and is of a flakier texture than scale from the products of combustion. A magnetic check of the material offers the most conclusive test; oxide scale is magnetic, and scale from the products of combustion is nonmagnetic.

Heater tubes that have been in service may become temper embrittled and have low ductility at ambient temperature. To avoid any possible damage, carbon and alloy steel heater tubes should have a minimum metal temperature of about 60°F (15.6°C) during hammer tests.

In certain cases, the hammer testing of tubes can lead to damage. Austenitic stainless steel tubes may suffer strong stress corrosion cracking at areas that are cold worked by hammering. Cast tubes and chromium alloy tubes should not be hammer tested.

SECTION 6—LIMITATIONS OF THICKNESS

6.1 General

Unless the limits of the degree of deterioration that may safely be tolerated are well known for the particular part being inspected, the inspection will lose considerable value. Two factors must be determined to evaluate a part: the rate at which the deterioration of the part is proceeding and the limits of safe deterioration of the part.

6.2 Heater Tubes

Methods of establishing minimum allowable thicknesses range from the highly complex to the simple. With the average heater, the operating pressure and temperature are known only for the heater inlet and outlet. The pressure and temperature at intermediate points must be determined by calculation, estimation, or installation of pressure gauges and thermocouples.

The metal temperature largely governs the working stress that should be allowed for a given tube material. For a given tube size and a given operating pressure, the thickness limit varies with the allowable working stress.

API Recommended Practice 530 gives extensive information on the calculation of required wall thicknesses of new tubes (carbon steel and alloy tubes) for petroleum refinery heaters. The procedures given are appropriate for designing tubes or checking existing tubes in both corrosive and noncorrosive services.

Many methods, including those involving tube skin thermocouples, infrared cameras, infrared pyrometers, and optical pyrometers, are available to determine the metal temperature of a tube. A simple method is to estimate the metal temperature from the operating fluid temperature and then adjust the temperature estimate based on the location of the tube in the heater—the skin temperatures on a tube closer to the flame or nearer the heater outlet will be hotter than one at the heater inlet.

Another way to determine the minimum allowable thickness of heater tubes is to use the inlet pressure and the outlet wall temperature in a simple empirical formula like the following equation:

$$t = \frac{PD}{2SE} \quad (1)$$

Where:

t = internal pressure design thickness.
P = internal design gauge pressure.
D = outside diameter of the pipe.
S = stress value for the material.
E = joint or quality factor.

This formula can also be found in ASME B31.3.

An additional way to determine the minimum allowable thickness of tubes is to calculate it for the actual pressure and metal temperature at the inlet and the outlet of the heater, using a simple empirical formula. If the difference between these thicknesses is great enough, the minimum thicknesses at various points can be interpolated between these values.

Under certain conditions, the methods described in the preceding text may result in a thickness that is too small for practical purposes. The minimum allowable thickness must be great enough to give the tube sufficient structural strength to prevent sagging between supports and to withstand upset

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operating conditions. For this reason, it is customary to add some amount based on judgment and experience to the calculated minimum allowable thickness and to use this greater thickness as the limit at which a tube should be replaced.

When determining the need to replace tubes with metal temperatures in the creep range, the amount of diametral creep is another factor that should be taken into account. The increase in tube diameter should be well within the range of creep that will not cause rupture. In wrought tubes only, it is common to limit the increase in diameter resulting from creep to 5 percent of the tube's original external diameter.

6.3 Heater Fittings

When establishing the minimum allowable thickness for heater fittings, the metal temperature of a fitting outside the fire zone of a heater is usually considered to be the same as the temperature of the fluid flowing through it. The metal temperature of a fitting inside the firebox is considered to be the same as that of the corresponding tubes. The allowable working stress value for fittings is determined in the same way as it is for tubes. An empirical formula like Equation 1 is generally used to calculate the minimum allowable thickness. Because of the complex shape of heater fittings, it is generally advisable to add a shape factor to the formula being used.

Fittings produced according to ASME B16.28 are intended for use at pressure ratings equal to 80 percent of those calculated for seamless pipe of the same size, nominal thickness, and equivalent material, in accordance with the rules established in ASME B31.1, B31 Guide, B31G, B31.2, B31.3, B31.4, B31.5, B31.8, B31.9, and B31.11. Fittings produced according to ASME B16.9 are pressure rated as calculated for straight seamless pipe in ASME B31.1, B31 Guide, B31G, B31.2, B31.3, B31.4, B31.5, B31.8, B31.9, and B31.11.

Because of stresses that may be set up by closing and holding members and by thermal expansion, the calculated allowable thickness may be too small to be practical. As with tubes, it is advisable to add some arbitrary thickness, based on judgment and experience, when setting the minimum thickness at which a heater fitting should be replaced.

When plugs are used in a heater fitting like plug-type or mule-ear fittings (see Figure 27) or when a sectional L is used in a sectional fitting (see Figure 27), the width of the seating surface in the fitting must be sufficient to prevent leakage. A width large enough to prevent leakage generally provides adequate strength against blowout, but a lesser width should never be used. The proper seating width required to prevent leakage can only be determined by experience. When there is no previous experience to be used as a guide, the best way to determine these limits is to wait until evidence of slight leakage is found and then set a limit at a point that is a little greater than that at which the slight leakage was evident.

6.4 Boiler Components

Because of the great number of variables affecting the limiting thickness and the variety of types, sizes, shapes, operating methods, and constructions of boilers, it is not possible in this recommended practice to present a set of precalculated minimum or retiring thicknesses. However, it may be quite feasible to prepare one for the boilers in a given refinery. Formulas for the thickness of drums, headers, and tubes are given in the ASME *Boiler and Pressure Vessel Code,* Sections I and IV. These formulas can be used as guides when repairs and replacements are needed.

SECTION 7—METHOD OF INSPECTION FOR FOUNDATIONS, SETTINGS, AND OTHER APPURTENANCES

7.1 Foundations

All foundations can be expected to settle to some extent. If the settling is evenly distributed and only to a small extent, little or no trouble may be experienced. If the settling is uneven or to a large extent, serious consequences may result. Whether even or uneven, any settlement in a foundation should be studied and, if the need is indicated, checked at frequent intervals by level measurements, which should be continued and plotted until the settlement practically ceases. When settlement is first noted, all pipe connections to the heater should be examined carefully to determine whether they are subject to serious strain and consequent high stress. If conditions warrant corrective measures, they should be taken immediately.

One of the main causes of the deterioration of foundation concrete is high temperature. This causes calcining, which is caused by the concrete's loss of water of hydration and leaves the concrete a weakened mass with very little cohesion. Calcining can easily be detected by chipping at the suspected area with a hammer. If calcining is present, the concrete will fall away as a powder with very little impact from the hammer.

Spalling is another form of concrete deterioration. This is caused by heat or an insufficient thickness of concrete over the reinforcement. The concrete cracks, and moisture can enter and attack the steel reinforcement. The products of corrosion build up and exert sufficient pressure against the

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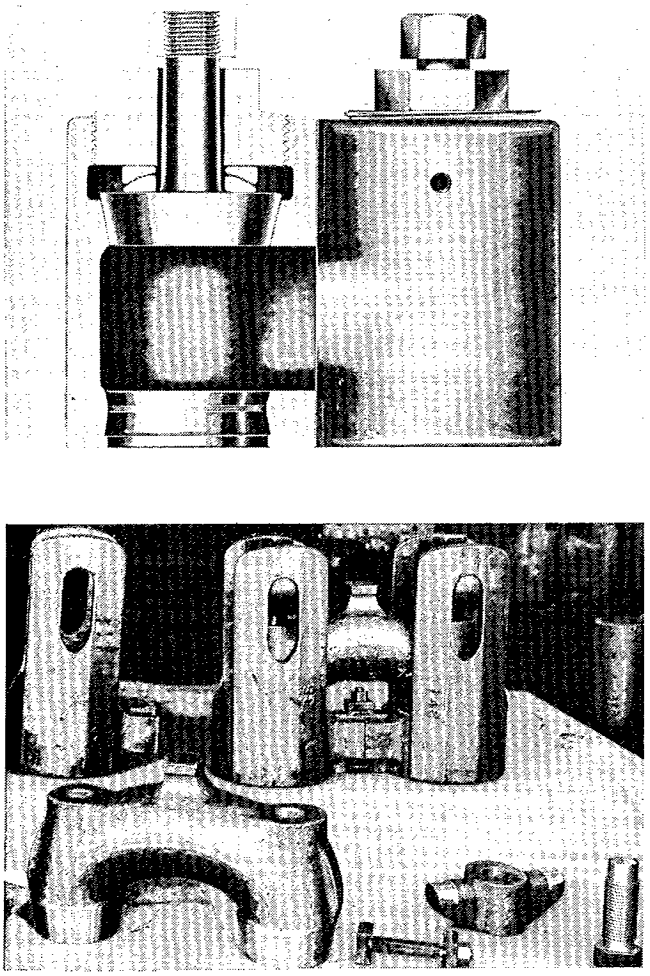


Figure 27—Types of Heater Fittings

concrete covering to cause it to flake or spall, exposing the reinforcement to further attack. Only a visual inspection is necessary to detect this form of deterioration.

7.2 Structural Supports

A visual inspection should be made of all load-carrying structural steel members to see whether deflection is observable. If bending is present in a column, it may be caused by overloading, overheating, or lateral forces applied to the column by the expansion of elements in the furnace. These potential causes should be sought, and the cause of the bending should be determined so that proper corrective measures can be taken.

If the bending is due to overloading, either the column should be reinforced by welding or riveting the necessary reinforcement to the column's web to reduce the unit stresses to a permissible value, or the column should be replaced with another one of suitable size. If the bending is caused by overheating, the column should be protected by insulation or a shield. If the bending is caused by expansion of elements in the furnace, provisions should be made to accommodate the expansion without stress on the column.

Beams and girders will deflect when loads are imposed on them. The deflection should be measured where it is greatest. The amount of deflection should be checked against that calculated for the load on the beam or girder. If the measured deflection is greater than the calculated deflection, overstressing is indicated. If the overstress is serious, the design should be investigated, and corrective measures should be taken.

If corrosion in structural steel members that bear loads directly is so great that the thickness lost is enough to weaken the part, the minimum cross-sectional areas should be measured carefully after the corroded part is cleaned thoroughly to permit the determination of the remaining sound metal. When the measurement has been obtained and the remaining sectional area has been determined, the section modulus should be calculated, the design should be checked to determine the stress. If the stress is sufficiently higher than the allowable stress, the weaker part should be reinforced or replaced. Useful design information, including information about allowable working stresses, can be found in AISC M015L and M016.

The connections between the columns and the beams and girders should be inspected visually. These connections may be made by riveting, bolting, or welding. For riveted or bolted construction, broken or loose rivets or bolts can be detected by striking the side of the rivet or bolt and by striking the plate. A movement of the rivet or bolt will indicate that it is loose or broken. Inspection of all connections is not warranted, but inspection should be made where corrosion is severe. If the connections are welded, corroded sections should be carefully visually inspected af-

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ter proper cleaning, and the effect of lost metal thickness should be determined.

7.3 Setting, Exterior, and Casing

The exposed parts of the setting should be inspected for signs of deterioration. All metal parts can be adequately inspected with a hammer and visual examination. If the exposed parts are painted, a visual inspection should be made to see whether the coating adheres tightly to all surfaces. Areas exposed by flaking or otherwise damaged should be cleaned and repainted. The casing should be inspected for thinning or perforation due to acidic flue-gas corrosion.

Stairways, walkways, and platforms should be checked to ensure that they have not been materially weakened as a result of corrosion. Heater header boxes should be inspected for warpage and improper functioning. Warpage or improper functioning of doors may allow rain or other moisture to enter. Header box warpage also allows excess air into the furnace, spending additional fuel. In some operations, particularly those with heaters that process light hydrocarbons, a sudden change in temperature due to leakage of header boxes can cause enough movement in fitting closures or rolls to loosen them.

Peepholes, access doors, and the like should be inspected visually to see that the fit is satisfactory and minimizes excess air ingress.

Explosion doors, if provided, should be inspected visually for corrosion of the hinges and the door itself and for warpage. Explosion doors should also be visually inspected to see whether there is proper seating contact between the door and the door frame, ensuring a reasonably tight joint. The doors should be manually lifted to check operability. To serve effectively, the doors should open with minimum resistance.

7.4 Refractory and Insulation

Most modern settings consist of structural steel framing with refractory lining or lightweight ceramic or blanket insulation on the walls and roof of the heater. The refractory may be backed up with brick or supported on steel members with heat-resistant hangers. The supporting brickwork and reinforced concrete and the clearance in the expansion joints should be examined for deterioration due to heat, open joints, excessive distortion, or debris. The inspection of refractory should consist of a visual examination for breakage, slagging, crumbling, and open joints. Leakage of the hot furnace gases through joints when the edges have crumbled or when the tile or insulating concrete has fallen out exposes the supporting steel to high metal temperatures, rapid oxidation, and corrosion. Leakage of hot furnace gases outward instead of air leakage inward may indicate improper draft conditions in the firebox. The supporting steelwork should be inspected thoroughly. Beams, hangers, and supports of any type that have been damaged by heat or show excessive distortion should be replaced. Any accessible insulation used on the exterior should be inspected.

7.5 Tube Supports

7.5.1 GENERAL

Tube sheets and tube supports should be examined to determine their physical condition and fitness for further service. Supports should be examined carefully for cracks, oxidation, and corrosion. If found to be unsound or weak, they should be reinforced or replaced.

7.5.2 STEAM/METHANE–REFORMING HEATERS

Tube support methods vary in steam/methane-reforming heaters. Some designs require full support from the top. In these designs the pigtail may be below the tube and unable to take any load from the catalyst-filled tube. Counterweights are often used and may support two or more tubes. The lever or pulley system must work as designed. Interference from tube flange bolts, slipping of supports off tube flanges, and other similar problems have led to pigtail failures.

Inadequate support also allows tube bending, which puts a bending moment on a pigtail that exits the tube from the side, thus causing localized high stress at the fitting on the tube or the outlet headers.

Outlet headers grow, usually from a center anchor point. Bottom tube supports on short pigtailed tubes must allow movement of the tube bottom to minimize stress on the pigtail. If the tube is designed for bottom movement, the upper tube supports must allow the tube to move at the bottom end. To prevent a pigtail bending moment, the furnace lining must not press on the tube. Loose bricks are often used to help close openings. The bricks must move freely if the tube presses on them.

If support springs are used, those that have been stretched should be replaced. A stretched spring cannot support a tube. When the tube is heated up after shutdown, the spring will no longer support it as designed.

7.6 Visual Inspection of Auxiliary Equipment

7.6.1 GENERAL

In addition to any external inspection of auxiliary equipment while the furnace is in operation, a close inspection should be made of each piece of equipment while the unit is out of operation. Indications of malfunctions noted during external inspections should be investigated, and any indicated repairs should be made. Since some parts wear out and fail without warning, manufacturers' catalogs and instructions should be reviewed so that all critical operating parts may be investigated.

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7.6.2 DAMPERS

Power-operated or manual dampers are provided on some but not all boilers for superheater, economizer, and boiler outlet-gas control. Damper blades, constructed of thin metal, are susceptible to oxidation and warpage due to overheating and should be inspected for such damage. Supporting brackets, driving rods, pins, and other devices should also be examined. The dampers should be operated and checked for binding closure, and freedom from obstructions should be ensured. Personnel, other than those working on damper operation, should not be permitted in the damper section while the dampers are in operation.

7.6.3 FORCED– AND INDUCED–DRAFT FANS

The bearing clearance and the condition of the babbit-bearing surfaces and of the antifriction bearings should be checked, and the shaft diameter should be measured at the bearing surface. The condition of the oil or grease should be checked, and the lubricant should be changed as required.

The general condition of the rotor and rotor blades should be checked, and loose blades should be fixed. Couplings should be examined, and the alignment of all parts should be inspected. If any parts are out of alignment, the cause should be determined and corrective action should be taken. Any dampers should be tested for ease of operation and freedom from obstruction.

Induced-draft fans are subject to erosion and corrosive attacks from ash particles and flue gas. In addition to the inspections discussed in preceding text, inspections of the rotor blades and casings should be made for corrosion, excessive thinning, and holes in the blades and casing. The shaft should be examined for corrosion due to dew-point condensation near the casing. Missing or faulty gasket seals around the shaft will allow the entry of cold air and lead to condensation and subsequent corrosion.

7.6.4 SOOT BLOWERS

Soot-blower parts should be inspected for proper alignment, warpage, and position. If soot blowers are out of position, the blower blast impinging on nearby tubes will eventually cause tube failure due to erosion. The blower, supporting hangers, and brackets should be examined visually for soundness and for excessive thinning from oxidation. Soot blowers for the high-temperature part of the boiler are sometimes made of high-chromium alloys that embrittle in service. If these are hammer tested too vigorously, they will crack.

Connection welds of supporting elements should be inspected for cracks. If the welds look cracked, a magnetic-particle inspection should be made. Packing glands and all operating parts of the rotating and retracting types of soot blowers should be examined for good working condition. Because of the potential difficulty of repacking soot blowers in service, repacking should be done during down periods if there is any evidence that repacking might be required.

7.6.5 AIR PREHEATERS

Air preheaters are subject to corrosion due to condensation during extended periods of downtime. Recuperative preheaters, both the tubular type and the plate type (see Figures 7 and 28), are subject to severe corrosion when the element temperature is at or near the dew point. The severe corrosion is particularly prevalent at the flue-gas outlet end. As much as possible of the recuperative-type preheaters should be inspected for corrosion. Usually, the conditions at the inlet and outlet ends will provide a good indication of what can be expected in the remainder of the preheater. It is not unusual to see extensive plugging of air preheaters when boilers are being fired with heavy oil or coal.

Perforated tubes should be replaced or plugged. It is sometimes necessary to remove fairly good tubes or plates to get to the bad ones. Good judgment and consideration for future replacements are important factors in selecting the most economical method for repairing tubes and plates.

Regenerative preheaters (see Figure 8) require a more extensive inspection than do recuperative preheaters. Usually, rotating elements must be removed to clean the preheater. This affords an opportunity for close inspection of all parts. In most classes of regenerative preheaters, the incoming air enters at the same end that the flue gases leave, thus cooling that layer of rotor segments first. Corrosion will generally start at this point because of condensation and proceed toward the other end of the unit. Most preheaters have two sections, and if corrosion at the flue-gas exit ends is not too severe, the sections can be reversed; otherwise, new sections should be provided.

Rotor seals should be examined for corrosion. They can also be mechanically damaged by falling material, by high-pressure steam or water from soot blowers, or by being stepped on by maintenance personnel.

Soot blowers for regenerative preheaters are quite different from those used in other parts of the boiler. Manufacturers' catalogs and drawings should be examined for points that require close inspection. Soot blowers should be inspected for deposits and leaky valves. Leaky valves and buildup of ash cause corrosion of nozzle tips, and subsequent malfunction of the blowers damages rotor seals and segments. Therefore, steam inlet valves should be inspected for tight shutoff, and drain valves should be inspected for correct operation.

7.6.6 BOILER BLOWDOWN EQUIPMENT

Valves should be inspected for tight shutoff. Piping should be checked for corrosion and leakage at all joints. Ultrasonic testing and hammer sounding are good methods of pipe in-

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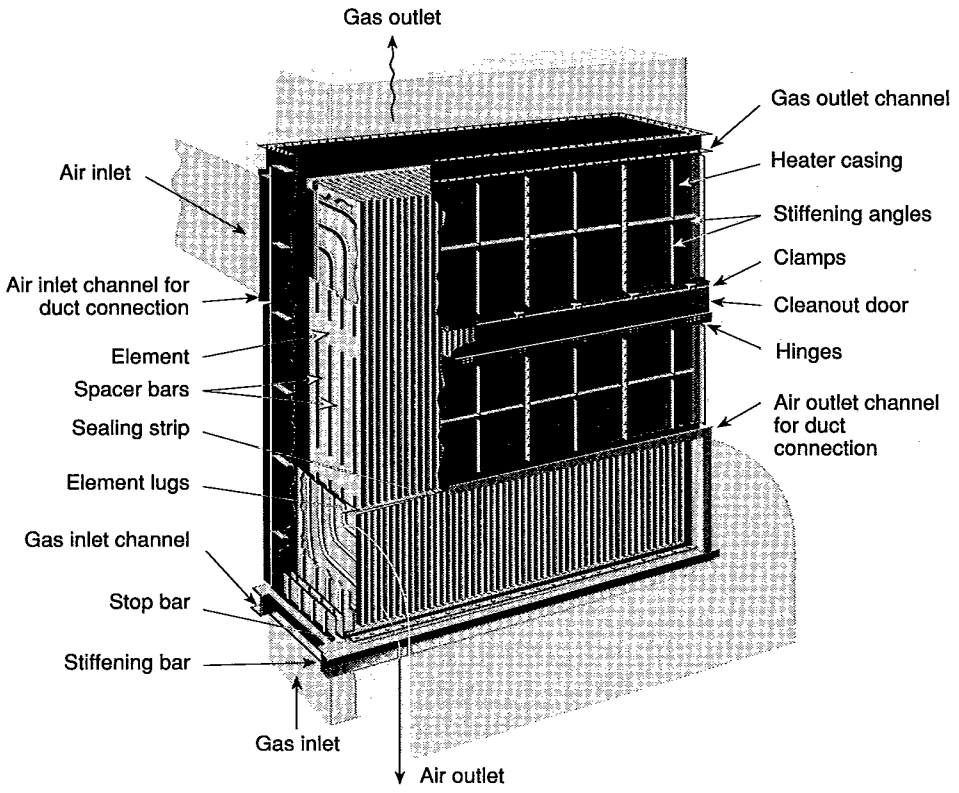


Figure 28—Plate-Type Air Preheater (Recuperative Type)

spection. Elbows and sharp bends are susceptible to erosion and should be examined for indications of thin walls and holes. Coolers should be inspected in the same manner as that described for heat exchangers in API Recommended Practice 572.

7.6.7 FUEL–HANDLING EQUIPMENT

7.6.7.1 General

Manufacturers' instructions, sketches, and drawings should be consulted before inspecting fuel-handling equipment.

7.6.7.2 Gas

Gas system equipment is not generally subjected to severe corrosion or wear and therefore does not require extensive inspection. This might not be true for boilers firing refinery fuel gas. The seats and packing of control valves, block valves, and bypass valves should be examined, and the valves should be checked for ease of operation and tight shutoff. Burner inspection will depend on the type of burner to be inspected. Usually, operating conditions will indicate the condition of burners. Malfunctioning may be due to fouled or cracked burners or burned burner tips. When the system contains a dry or knockout drum, planning is advisable so that the drum can be removed from service for inspection as required.

7.6.7.3 Fuel–Oil Pumps

Fuel-oil pumps should be inspected to ensure that they meet the standards called for when originally purchased (refer to applicable API standards). Fuel-oil heaters should be inspected as indicated in API Recommended Practice 572. Valves and burners should be inspected as indicated in the preceding text for gas equipment valves and burners. When the fuel contains corrosive products, all items should be examined for evidence of corrosion.

7.6.8 BURNERS

Burners should be visually inspected while the unit is in operation, and any necessary adjustments should be brought to the attention of the person responsible. If the burner requires mechanical repairs that cannot be performed while

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the unit is in operation, the defective burner should be removed and replaced immediately with a spare burner. This replacement of defective burners during unit operation is an important factor in the maintenance of uniform firing conditions. Poor firing from unbalanced burners can cause serious deterioration of the heating elements and setting. Burners and burner blocks should also be inspected when the unit is shut down.

SECTION 8—STACKS

8.1 Flue-Gas Stacks

An external visual inspection should be made of brick, concrete, and steel stacks for conditions that may weaken these structures. Field glasses will be helpful in making inspections of high stacks because they will enable any defects to be observed fairly well from the ground. Brick stacks should be inspected for cracks and for the condition of mortar joints to determine the effect of weathering. Concrete stacks should be inspected for cracks and spalling that may expose the steel reinforcement. Steel stacks should be inspected externally for the condition of painted surfaces, signs of oxidation, and thinning or perforation due to corrosion by acidic flue gases.

In many cases, cracks in brick and concrete stacks are due to insufficient thickness of the internal insulation or to internal secondary combustion. These potential causes of cracks should be kept in mind when inspecting the interior of stacks.

The linings of all stacks should be inspected for cracks, wear, and structural soundness.

While stacks are in service, an external thermographic examination can be made that will show hot spots, which indicate failure of the internal liner.

When liquid fuels are burned, soot accumulates in the base of the stack and must be removed occasionally. During the internal inspection, the amount of soot and ash should be noted, and whether they need to be removed should be decided. The inside of steel stacks should be inspected for corrosion or cracking due to condensation of acidic flue gases. Areas at or adjacent to welds are most susceptible to stress corrosion cracking.

Steel stacks in heater, boiler furnace, flare, and blowdown services should be inspected and checked for wall thickness at time intervals that are warranted by experience. In addition to the thickness determination, a thorough hammer inspection should be made of the entire stack, with particular attention paid to the seams, adjacent areas, and areas adjoining any stiffening rings, lugs, nozzles, and the like, which may act as cooling fins to cause condensation of gases and localized corrosion. The minimum allowable thickness at which repairs will be made should be definitely established for such structures. The best practice is usually to establish these thicknesses on the same basis as was used in the original design for the structure (see Figures 29 and 30).

Bolts at the base flange and at elevated sections should be checked periodically for loosening and breakage. Elevated flanged connections that are installed for the purposes of field erection should be seal welded internally to prevent the escape of corrosive flue gases, which accelerate bolt failure. In the case of derrick-type flare stacks, the structure itself should be completely inspected. Careful attention should be given to the foundations and anchor bolts. Most derricks are assembled by welding or bolting. Bolts should be checked for looseness and corrosion. If looseness is found, the shank of the bolt should be checked for abrasion from the movement of structural members. The flare-stack roller guides and guide arms should be checked for alignment and operability and should be realigned or freed if necessary. Ladders, platforms, and all structural members should be checked for atmospheric corrosion to determine whether any section is approaching the minimum allowable thickness.

The guy lines to guyed steel stacks should be inspected visually for corrosion. Connections to the deadmen at the bottom and to the stack at the top are especially subject to corrosion because of the possibility of moisture settling and being retained around these connections. It is impractical to completely inspect the guy lines between the deadmen and the top of the stack. For this reason, it is considered good practice to replace the guy lines at some safe interval that can be determined after the results of several inspections have been analyzed.

The stack painters' trolley and cable should be inspected visually for corrosion or mechanical damage before being used and before being returned to storage. The condition of the connections at the top of the pulley and of the trolley ring and its connections to the stack should be determined carefully.

Lightning rods on stacks and their grounding cables should be inspected visually to see that they are secured and unbroken. The ground rod should be inspected visually to see that it is firmly attached to the cable and that it extends to a ground depth sufficient to provide an electrical resistance of not more than 25 ohms. This should be checked periodically, particularly in dry weather.

The ladders on steel, concrete, and brick stacks should be inspected visually for corrosion and should be tested physically by applying test weights in excess of those that may be imposed by the personnel using them.

The caps on radial brick and concrete stacks sometimes become damaged, causing loose brick to fall or the reinforcing steel to be exposed. Stack caps should be inspected visually so that any necessary repairs can be made, thereby

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eliminating a hazard from falling bricks and preventing damage to steel reinforcement.

Note: Extreme care should be taken when climbing a stack for inspection if the stack cap has been struck by lightning.

8.2 Flare Stacks

Flare stacks are used to burn excess gas released from operating units under the following conditions:

a. When surges in pressure occur.
b. When relief valves operate.
c. During purging of equipment.

Flare stacks are usually constructed of steel and are erected to a height well above that of any surrounding equipment. They are provided with a gas pilot light at the top to ignite any gas as it is released. A knockout drum is generally provided in the gas line to the flare to remove condensed hydrocarbons. Flare stacks may be self-supporting, guyed, or supported within a derrick-type structure.

8.3 Blowdown Stacks

Blowdown stacks are used to release gas, volatile liquids, or volatile vapor from certain equipment during some phase of the operating cycle or in an emergency when it becomes necessary to empty a unit quickly.

Blowdown stacks are usually constructed of steel. They are equipped with a knockout drum through which the blowdown from the unit must pass before any gas or vapor enters the stack. The purpose of this vessel is to knock out any liquid that may be present and allow only the gas or vapor to enter the stack.

Blowdown stacks are usually equipped with internal water sprays to cool vapors. Stacks of this type may be either self-supporting or guyed, and their height should be above that of adjacent equipment. Figure 30 shows a typical blowdown stack.

SECTION 9—METHOD OF REPAIRS

9.1 Heaters

In general, repairs to heaters entail repairing or replacing parts that have become weakened or damaged. Many relatively good heater tubes have to be condemned and removed from service because of excessive bowing and sagging or defective fittings. However, if the metal-wall thickness of a tube with excessive bowing and sagging or defective fittings is sufficient, the tube can be reworked or salvaged. Reworking generally entails cleaning, straightening, and welding on a stub to restore the tube to a standard length.

When a tube is to be salvaged by welding a new piece onto it, special attention should be given to the alignment and the uniformity of thickness at the ends of the section to be joined. If necessary, the ends should be tapered on the inside to obtain sufficient uniformity. When the weld is completed, a thorough inspection should be made. If the inspection shows improper alignment or poor or excessive metal penetration at the weld, the tube should be rejected for service until the proper repairs have been made. The welding operation, including preheating, welding technique, and post-heating, should be performed according to the best approved methods or practices recommended for each particular type of material.

There are several types of repairs that can be made to heater fittings, both when they are in the heater and when they are removed. The most common repair consists of welding small cracks that develop in the fitting. Solid fittings with horseshoe-type holding sections that have become stretched and misshapen can be reconditioned by heating and reshaping with a homemade forming tool. The horseshoe section is sometimes reinforced with a flat vertical bar welded on each end of the section.

Plugs or U-bend seats in the fitting that have been damaged or deformed can be reworked by machining and grinding. Seat liners or oversized plugs or U bends may be used. In some cases, the tube seat in a fitting can be reconditioned simply by the installation of a tube seat liner, without prior machining or grinding.

The repair of furnace settings, ducts, stacks, and the like usually involves routine patching or replacement. It is sometimes possible, however, to extend the service life of an unlined steel stack that has partially corroded by installing a monolithic insulating liner in the corroded area.

9.2 Boilers

9.2.1 GENERAL

The repair and maintenance of boilers are not described in this recommended practice as they seldom are the function of the inspector. Repairs to pressure parts must be made in accordance with applicable regulations. These regulations may require approval of the proposed method of repair by a commissioned state, provincial, or insurance inspector.

After pressure parts are repaired, the work should be inspected and tested in accordance with the applicable code and law.

9.2.2 TESTING OF BOILERS

When boilers are first built, they are tested in accordance with the standard to which they were constructed. When

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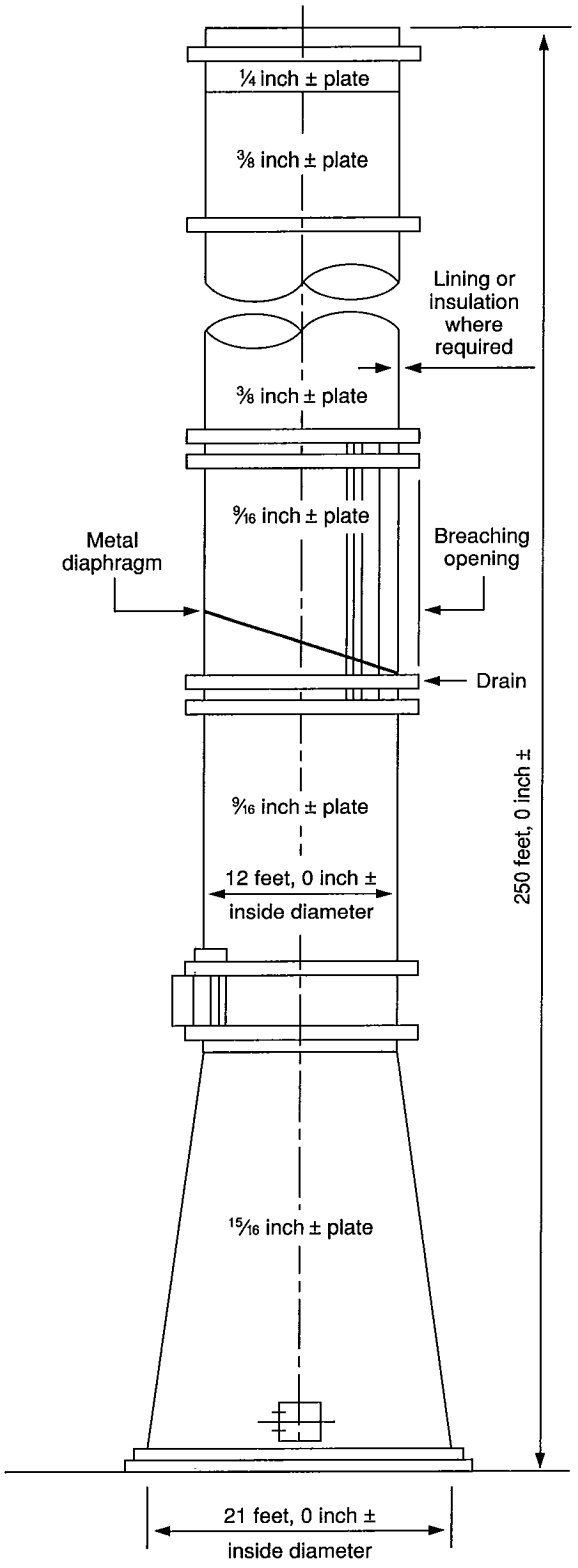


Figure 29—Self-Supporting Steel Stack

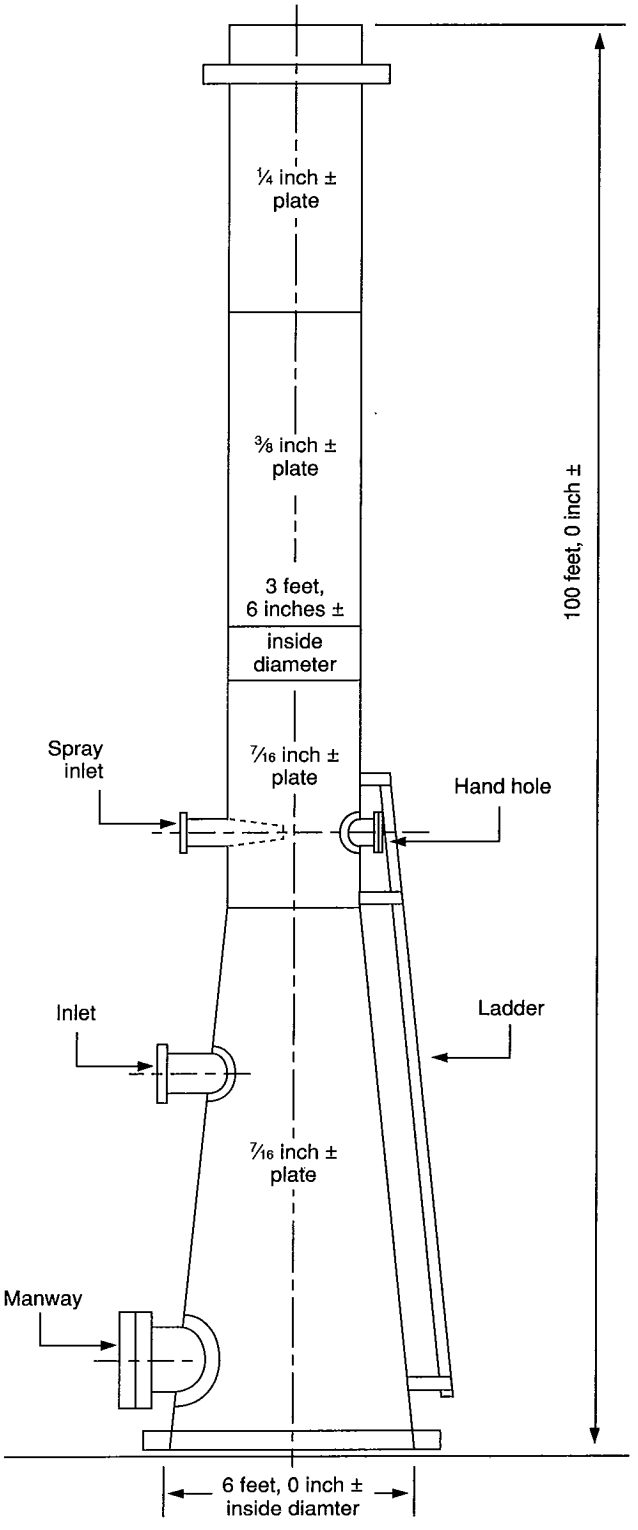


Figure 30—Blowdown Stack

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practical, the same methods should be used to inspect and test repair work. After a routine internal and external inspection, boilers are not usually tested to the original hydrostatic test pressure that was required before they were put into operation. However, it is good practice to test them for tightness by applying a hydrostatic test pressure that is near the normal working pressure. Repeated, unnecessary pressure testing at 1.50 times the maximum allowable working pressure may be harmful to equipment and is not recommended unless required by governmental or insurance agencies, as a result of a major repair or alteration to the boiler pressure parts, or for some other practical strength or safety reason. Close inspections of boiler pressure parts should never be carried out at 1.50 times the maximum allowable working pressure. For personnel safety, close inspections of boiler pressure parts should be conducted at 1.00–1.25 times the maximum allowable working pressure. Roll-joint and rivet-seam leaks evident at higher pressures may not require repair unless such leaks occur at the maximum safety-valve set pressure.

In this section, the word *testing* applies only to the process of filling the boiler with water at the appropriate pressure and thereby testing the strength of the boiler and its tightness against leaks. It is important that all air be vented from the boiler so that it is completely full of water before the pressure is raised. In no case should the water temperature be less than 70°F (21°C). It is preferable that water temperature be no more than 50°F (10°C) higher than the boiler metal temperature.

SECTION 10—RECORDS AND REPORTS

10.1 General

The importance of keeping complete records cannot be overemphasized. Inspection records form the basis for determining reliability and establishing a preventive maintenance program. With good, complete records, it is usually possible to predict when repairs and replacements will be needed. This helps prevent emergency out-of-service time. It also saves time by allowing personnel and materials to be scheduled before a shutdown. Records can also be used for reference in preparing specifications for new equipment.

10.2 Heater Records

There are certain basic inspection data on heater tubes and fittings that all inspection organizations consider necessary. It is important to measure and record the thickness of new tubes when they are installed. If this is not done, the first inspection period may not accurately reflect actual corrosion rates. If the installed thicknesses of the tubes are not available at the time of the first inspection, corrosion loss is determined on the assumption that the wall thicknesses of the new tubes were exactly as specified on the purchase order. This is not always true, and hence an error in the calculation of corrosion rate may result.

The types of forms that may be used for recording the necessary information vary widely among companies.

10.3 Boiler Records

Separate records should be kept for each boiler. A complete boiler record file should consist of at least three types of records: construction records, field notes, and historical records. Construction records should consist of prints, specifications, design data, all available results of material analyses and tests, and any other information relative to the construction or repair of the boiler. Field notes should consist of records made in the field either on prepared forms or in a field notebook. These notes should include all measurements taken, the conditions of all parts inspected, and a record of all repairs. A complete description of any unusual conditions encountered should also be kept in the field notes. Historical records should include all data accumulated for a boiler since the time of its construction. All measurements, repairs, and replacements are recorded in this section. Service conditions, records of any experiments with insulation, and firing rates should be recorded in the history section. Copies of all inspection reports should be kept as part of the historical record.

10.4 Reports

Inspection reports should be clear and complete. All unusual conditions observed should be reported fully, since what seem to be insignificant details may prove to be of importance in the future. When necessary, sketches, diagrams, and photographs should be incorporated in the report. There should be no unnecessary delay between the inspection and the submission of the report. Sample reports are shown in Appendixes A and B.

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APPENDIX A—SAMPLE RECORDS FOR HEATER TUBES AND FITTINGS

This appendix reproduces samples of the records kept by a company on the tubes and fittings of its heater. All of these records are used as field records, office records, and completed forms included in the report covering the inspection of the heater.

The *tube layout drawing* shows the actual arrangement of tubes and fittings in the heater. The flow through the heater is also noted. Tubes removed from the heater during the inspection and tubes approaching the minimum allowable thickness for service can be noted by a special color scheme.

The *tube inspection record* shows the history of all tubes in a heater on the date the current inspection is completed and the heater is ready to return to operation.

The *tube inspection record (record of tubes calipered)* is used to record the tube-calipering measurements taken during the current inspection. The figures set in roman type on the top half of each block are the measurements taken during the previous inspection. The figures set in italic type on the bottom half are the measurements taken during the current inspection. The two-digit figures to the right of the inside diameter measurements denote the change in inside diameter from the previous inspection and equal twice the corrosion rate for the interval between the two inspections. Once the report has been prepared, an extra copy should be made of this record and used as a field work sheet during the next inspection.

The *tube inspection record (instrument caliperings)* is used to record tube thickness measurements taken by radiography or with ultrasonic or radiation-type instruments.

The *tube renewal record* is used to record information on all of the tubes renewed during the interval between the completion of the previous inspection and the completion of the current inspection. It quickly shows the location of the tubes renewed and—of major importance—why the tubes were renewed and how long the tubes had been in service. This record is especially valuable when tube life and what tube material is best suited for the particular service are considered.

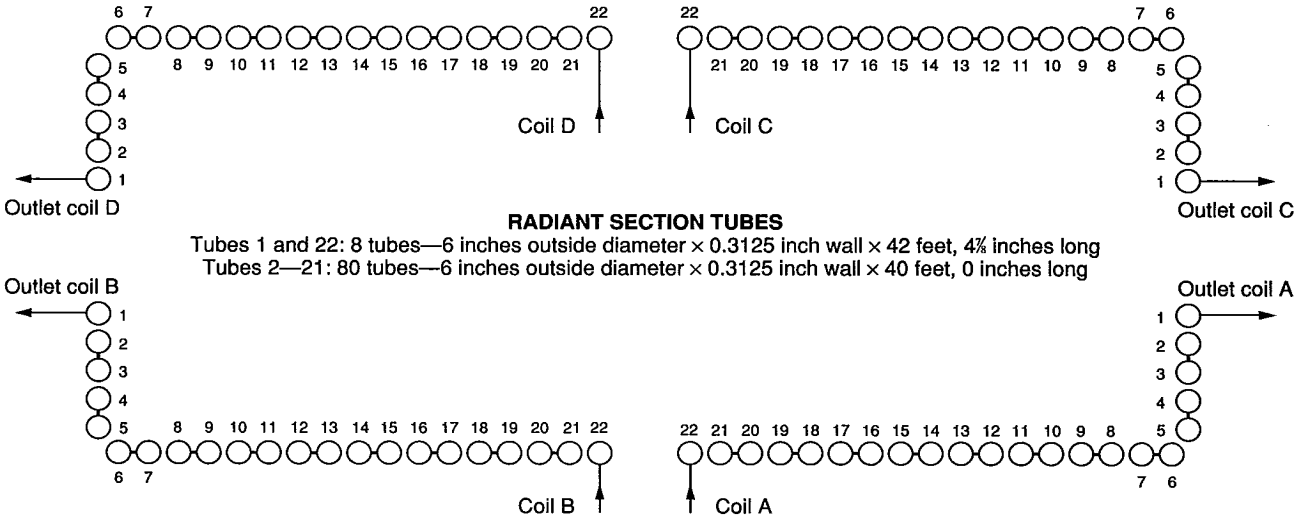
The *field work and record sheet (tube rolling data)* is used to record data necessary for the tube rolling operation.

The *record of heater fitting inspection and replacement* is primarily a reference record for heater fittings and shows where the various types of fittings should be checked for thickness. It contains a table for recording the actual outside diameters of a fitting at the various sections. Each point number on a sketch corresponds to a section of a fitting and not to a particular point on the fitting.

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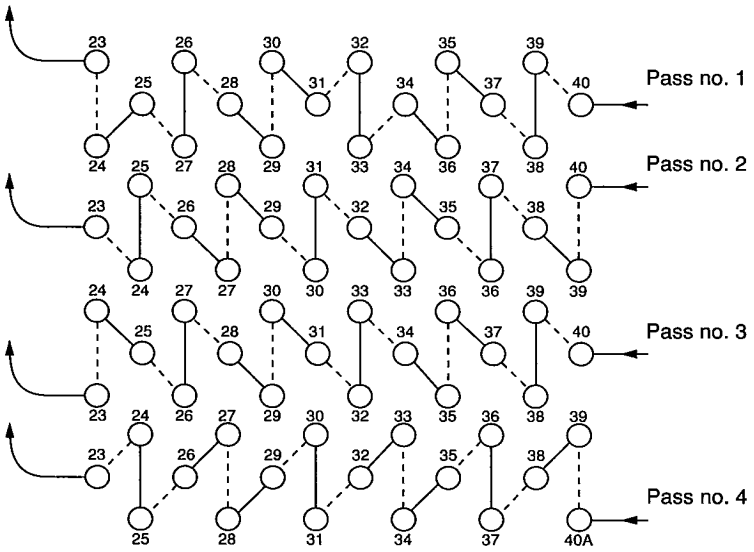
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SAMPLE TUBE LAYOUT DRAWING
(ATMOSPHERIC HEATER—CRUDE OIL PIPE STILL)



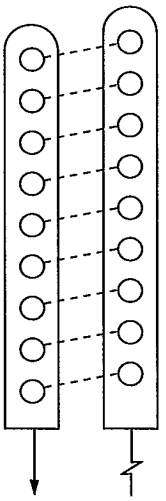
RADIANT SECTION TUBES

Tubes 1 and 22: 8 tubes—6 inches outside diameter × 0.3125 inch wall × 42 feet, 4⅝ inches long
Tubes 2—21: 80 tubes—6 inches outside diameter × 0.3125 inch wall × 40 feet, 0 inches long



CONVECTION SECTION TUBES

Tubes 23: 4 tubes—5 inches outside diameter × 0.3125 inch wall × 42 feet, 9⅛ inches long
Tubes 40: 3 tubes—5 inches outside diameter × 0.3125 inch wall × 40 feet, 0 inches long
Tubes 40A: 1 tubes—5 inches outside diameter × 0.3125 inch wall × 40 feet, 7½ inches long
Tubes 24–39: 64 tubes—5 inches outside diameter × 0.3125 inch wall × 40 feet, 6⅝ inches long



SUPERHEATER

18 tubes—
3 inches outside diameter ×
0.250 inch wall ×
1⅝ inches long

Notes:

1. A copy of this diagram is to be sent in with the tube inspection record after each periodic inspection and test.
2. Color in red all the tubes that are approaching minimum thickness at the time of inspection.
3. A copy of this diagram is to be sent in with the tube renewal record only when the arrangement of the tubes in the heater has been changed.
4. Tubes that are shown in this diagram but are not in the heater or in service are to be crossed out.
5. Tubes that are in the heater but are not shown in the diagram are to be shown in their relative locations and given the same number as adjacent tubes with the suffix "A."
6. The field is to indicate the actual flow when it differs from the flow shown on the diagram.

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SAMPLE TUBE INSPECTION RECORD (HISTORY OF ALL TUBES)

PLANT _____
UNIT _____
TUBE LAYOUT DRAWING _____
PAGE 1 OF 2
DATE _____

TUBE NO.	DATE INSTALLED	TUBE	
		MATERIAL	ORIGINAL OUTSIDE AND INSIDE DIAMETER (INCHES)
Economizer			
1-126	*2/24/67*	*1*	*3.5 × 2.7*
Preheater			
1	*12/16/70*	*2*	*4.5 × 3.5*
2	*2/29/72*	*1*	*4.5 × 3.5*
3-8	*4/17/70*	*2*	*4.5 × 3.5*
9	*9/19/70*	*2*	*4.5 × 3.5*
23	*9/19/70*	*2*	*4.5 × 3.5*
24-82	*4/17/70*	*2*	*4.5 × 3.5*
83	*4/17/70*	*1*	*3.5 × 2.7*
84-92	*10/26/72*	*1*	*3.5 × 2.7*
93-94	*4/24/72*	*1*	*3.5 × 2.7*
95-102	*10/26/72*	*1*	*3.5 × 2.7*
103-104	*4/17/70*	*1*	*3.5 × 2.7*
105-106	*10/26/72*	*1*	*3.5 × 2.7*
107-114	*2/24/67*	*1*	*3.5 × 2.7*
Side wall			
1	*7/8/71*	*1*	*4.5 × 3.5*
2	*1/15/72*	*1*	*4.5 × 3.5*
3	*12/17/71*	*1*	*4.5 × 3.5*
4	*7/23/72*	*2*	*4.5 × 3.5*
5	*1/4/72*	*1*	*4.5 × 3.5*
6	*7/31/72*	*2*	*4.5 × 3.5*
7	*1/4/72*	*1*	*4.5 × 3.5*
8	*7/23/72*	*2*	*4.5 × 3.5*
9	*4/24/72*	*2*	*4.5 × 3.5*
10	*7/17/71*	*1*	*4.5 × 3.5*
11	*4/24/72*	*2*	*4.5 × 3.5*
12	*1/10/69*	*1*	*4.5 × 3.5*
13	*4/27/72*	*2*	*4.5 × 3.5*
14-15	*1/10/60*	*1*	*4.5 × 3.5*
16-25	*4/24/72*	*2*	*4.5 × 3.5*
26	*1/22/72*	*2*	*4.5 × 3.5*
27	*1/29/72*	*2*	*4.5 × 3.5*
28	*1/10/69*	*1*	*4.5 × 3.5*
29	*7/23/72*	*2*	*4.5 × 3.5*
30	*1/15/72*	*1*	*4.5 × 3.5*
31	*7/16/72*	*2*	*4.5 × 3.5*
32	*1/15/82*	*1*	*4.5 × 3.5*

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SAMPLE TUBE INSPECTION RECORD (HISTORY OF ALL TUBES)

PLANT _____
UNIT _____
TUBE LAYOUT DRAWING _____
PAGE 2 OF 2
DATE _____

TUBE NO.	DATE INSTALLED	TUBE	
		MATERIAL	ORIGINAL OUTSIDE AND INSIDE DIAMETER (INCHES)
Side wall			
33	*4/27/73*	*2*	*4.5 × 3.5*
34	*12/26/72*	*2*	*4.5 × 3.5*
35	*4/5/73*	*2*	*4.5 × 3.5*

Notes:

Group tubes under headings such as *preheater, side wall, vertical, roof,* and *economizer.* Consecutive tubes may be grouped.

Kind of Steel:

1: Plain C	5: 9Cr-1.5Mo	9:
2: 4–6Cr	6: 14Cr	10:
3: 2Cr-0.5Mo	7: 18Cr-8Ni	11:
4: 4–6Cr-0.5Mo	8:	12:

Method for Reporting Welded Tubes:

1-1 for welded C steel.

2-2 for welded 4–6Cr steel.

7-2 for 18Cr-8Ni steel welded to 4–6Cr steel.

Method for Reporting Upset–End Tubes:

The symbol denoting the kind of steel precedes *U* as follows: *1U, 5U, 7U.*

Method for Reporting Tubes With Tube–End Liners:

The symbol denoting the kind of steel precedes *L* as follows: *2L, 4L.*

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SAMPLE TUBE INSPECTION RECORD (RECORD OF TUBES CALIPERED)

PLANT _____
UNIT _____
DATE _____
SHEET NO. _____

TUBE NO.	INSIDE DIAMETER IN ROLL (INCHES)				INSIDE DIAMETER IN BACK OF ROLL (INCHES)			
	TOP OR FRONT		BOTTOM OR REAR		TOP OR FRONT		BOTTOM OR REAR	
Economizer								
1	3.69		3.70		3.50		3.51	
	3.72	*0.03*	*3.72*	*0.02*	*3.51*	*0.01*	*3.51*	*0.00*
6	4.02	0.03	4.08	0.02	3.96	0.02	3.95	0.03
	4.06	*0.04*	*4.11*	*0.03*	*4.00*	*0.04*	*4.00*	*0.05*
20	4.10	0.05	4.11	0.04	4.01	0.03	4.00	0.02
	4.14	*0.04*	*4.16*	*0.05*	*4.03*	*0.02*	*4.02*	*0.02*
80	3.98	0.06	4.00	0.05	3.91	0.04	3.98	0.03
	4.05	*0.07*	*4.04*	*0.04*	*3.97*	*0.06*	*3.95*	*0.06*
Vertical								
1	4.48	0.04	4.50	0.03	4.32	0.02	4.36	0.04
	4.54	*0.06*	*4.54*	*0.04*	*4.35*	*0.03*	*4.41*	*0.05*
12	3.79		3.76		3.50		3.50	
	3.81	*0.02*	*3.80*	*0.04*	*3.52*	*0.02*	*3.53*	*0.03*
18	3.98	0.05	4.00	0.04	3.75	0.03	3.70	0.02
	4.06	*0.08*	*4.06*	*0.06*	*3.79*	*0.04*	*3.74*	*0.04*
49	3.87	0.04	3.90	0.05	3.61	0.02	3.59	0.04
	3.92	*0.05*	*3.94*	*0.04*	*3.62*	*0.01*	*3.63*	*0.04*

Note: Figures set in roman type refer to the previous inside diameter and change. Figures set in italic type refer to the current measured inside diameter and change. (When an inspection report is made, a copy of this form is to be saved for use as a field work sheet at the next inspection.)

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SAMPLE TUBE INSPECTION RECORD (INSTRUMENT CALIPERINGS)

PLANT _____
UNIT _____
SECTION _____
DATE _____
SHEET NO. _____

TUBE NO.	THICKNESS MEASUREMENTS (INCHES)					
	TOP OR FRONT		MIDDLE		BOTTOM OR REAR	

SAMPLE TUBE RENEWAL RECORD

PLANT _____
BATTERY _____
TUBE LAYOUT DRAWING _____
DATE _____

TUBE NO.	DATE INSTALLED	TUBE REMOVED		INSIDE DIAMETER IN ROLL (INCHES)		INSIDE DIAMETER IN BACK OF ROLL (INCHES)		CAUSE OF REMOVAL	DATE OF RENEWAL	NEW TUBE	
		MATERIAL	ORIGINAL OUTSIDE AND INSIDE DIAMETER (INCHES)	TOP OR FRONT	BOTTOM OR REAR	TOP OR FRONT	BOTTOM OR REAR			MATERIAL	ORIGINAL OUTSIDE AND INSIDE DIAMETER (INCHES)
Econo-mizer											
33	*11/4/70*	*1*	*4.5 × 3.5*	*3.70*	*3.75*	*4.06*	*3.98*	*D*	*6/15/73*	*1*	*4.5 × 3.5*
34	*11/4/70*	*1*	*4.5 × 3.5*	*3.72*	*3.78*	*4.00*	*4.08*	*D*	*6/15/73*	*1*	*4.5 × 3.5*
Vertical section											
8	*3/31/70*	*2*	*4.5 × 3.5*	*3.70*	*3.72*	*3.65*	*3.68*	*A*	*6/20/73*	*2*	*4.5 × 3.5*
9	*3/31/70*	*2*	*4.5 × 3.5*	*3.70*	*3.75*	*3.62*	*3.65*	*B*	*6/20/73*	*2*	*4.5 × 3.5*

Notes:

Group tubes under headings such as *preheater, side wall, vertical, roof,* and *economizer.* When tubes are renewed, this form is to be filled out and sent in as a monthly report or as a periodic inspection and test report. Calipering reported as *inside diameter in roll* is to be taken within 5 inches of each end of the tube. All tubes removed for any reason shall be shown and reported. Use two or more sheets of this form as necessary to cover all of the tubes renewed.

Kind of Steel:

1: Plain C	5: 9Cr-1.5Mo	9:
2: 4–6Cr	6: 14Cr	10:
3: 2Cr-0.5Mo	7: 18Cr-8Ni	11:
4: 4–6Cr-0.5Mo	8:	12:

Method for Reporting Welded Tubes:

1-1 for welded C steel.

2-2 for welded 4–6Cr steel.

7-2 for 18Cr-8Ni steel welded to 4–6Cr steel.

Method for Reporting Upset–End Tubes:

The symbol denoting the kind of steel precedes *U* as follows: *1U, 5U, 7U.*

Method for Reporting Tubes With Tube–End Liners:

The symbol denoting the kind of steel precedes *L* as follows: *2L, 4L.*

Cause of Removal:

A: Split tube	D: Thin tube
B: Burned due to split tube	E: Other causes
C: Bulged in operation	F: Burned in operation

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SAMPLE FIELD WORK AND RECORD SHEET (TUBE ROLLING DATA)

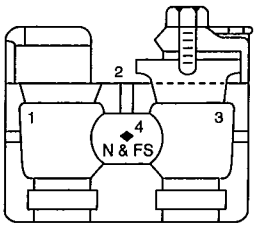
TYPE OF UNIT _____
UNIT NO. _____
PLANT _____
DATE _____

TUBE NO.	MATERIAL	FRONT OR TOP DIMENSIONS (INCHES)						REAR OR BOTTOM DIMENSIONS (INCHES)					
		INSIDE DIAMETER OF TUBE HOLE	TUBE END			INSIDE DIAMETER IN ROLL		INSIDE DIAMETER OF TUBE HOLE	TUBE END			INSIDE DIAMETER IN ROLL	
			OUTSIDE DIAMETER	INSIDE DIAMETER					OUTSIDE DIAMETER	INSIDE DIAMETER			
				IN ROLL	IN BACK OF ROLL	REQUIRED	ACTUAL			IN ROLL	IN BACK OF ROLL	REQUIRED	ACTUAL
Side wall													
8	*1*	*4.54*	*4.50*	*3.50*	*3.50*	*3.70*	*3.69*	*4.53*	*4.51*	*3.51*	*3.51*	*3.69*	*3.69*
10	*3*	*4.58*	*4.51*	*3.51*	*3.50*	*3.74*	*3.74*	*4.57*	*4.50*	*3.50*	*3.50*	*3.73*	*3.75*
19	*2*	*4.56*	*4.48*	*3.52*	*3.52*	*3.76*	*3.77*	*4.55*	*4.48*	*3.48*	*3.49*	*3.71*	*3.72*
26	*4U*	*4.53*	*4.50*	*3.34*	*3.49*	*3.55*	*3.56*	*4.53*	*4.50*	*3.35*	*3.50*	*3.56*	*3.55*
Preheater													
2	*1U*	*4.53*	*4.50*	*3.34*	*3.49*	*3.55*	*3.56*	*4.54*	*4.50*	*3.33*	*3.50*	*3.55*	*3.54*
8	*3U*	*4.54*	*4.51*	*3.35*	*3.48*	*3.56*	*3.56*	*4.56*	*4.49*	*3.34*	*3.48*	*3.60*	*3.61*
9	*4U*	*4.56*	*4.50*	*3.32*	*3.50*	*3.56*	*3.57*	*4.55*	*4.51*	*3.33*	*3.51*	*3.55*	*3.56*
10	*4*	*4.53*	*4.48*	*3.36*	*3.50*	*3.62*	*3.63*	*4.53*	*4.49*	*3.35*	*3.50*	*3.57*	*3.57*
85	*1-1*	*3.54*	*3.50*	*2.70*	*2.85*	*2.88*	*2.88*	*3.55*	*3.49*	*2.69*	*2.86*	*2.89*	*2.90*
87	*1U*	*3.55*	*3.48*	*2.56*	*2.69*	*2.78*	*2.79*	*3.56*	*3.50*	*2.54*	*2.68*	*2.76*	*2.77*
88	*2-2*	*3.56*	*3.51*	*2.69*	*2.75*	*2.88*	*2.89*	*3.54*	*3.50*	*2.70*	*2.79*	*2.88*	*2.89*
90	*4-2*	*3.54*	*3.50*	*2.70*	*2.84*	*2.88*	*2.87*	*3.54*	*3.51*	*2.71*	*2.80*	*2.88*	*2.87*

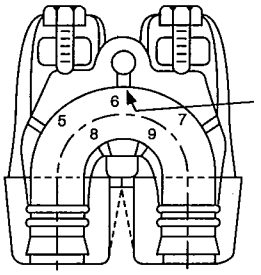
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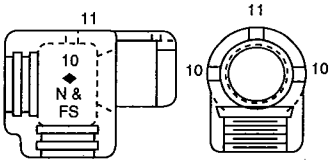
SAMPLE RECORD OF HEATER FITTING INSPECTION AND REPLACEMENT



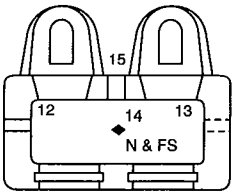
A. Cast steel junction box



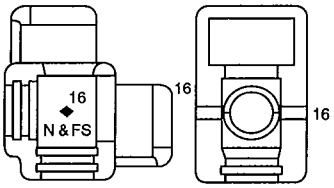
B. Streamlined return bend with U section bend



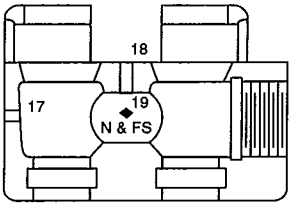
C. Cast steel terminal fitting—1 hole



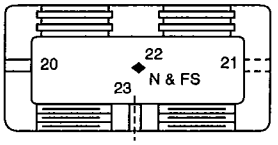
D. Cast steel junction box



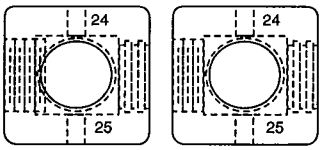
E. Cast steel corner fitting



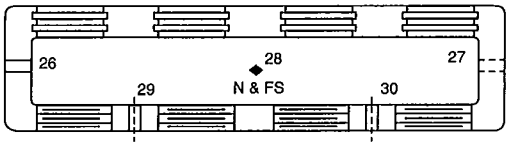
F. Cast steel terminal fitting—2 holes



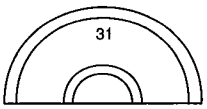
G. Cast steel return header—2 holes



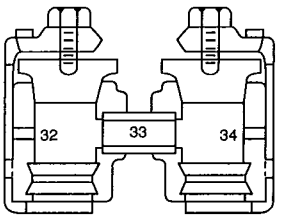
H. Forged box Ls



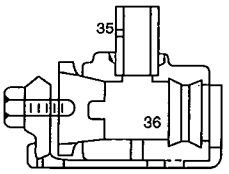
J. Cast steel return header—3 or 4 holes



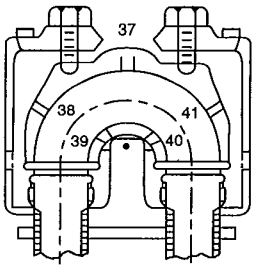
K. Steel return bend



L. Cast steel junction box



M. Cast steel terminal fitting



N. Streamlined return bend with U section bend

Notes:

1. The numbers shown on the sketches representing the sections of a fitting, not individual points.
2. The fitting number shall correspond to the tube number.
3. The symbols used to denote fitting material shall be the same as those used for tubes.
4. The average actual outside diameter at various sections of all sizes and types of fittings on the heater shall be recorded in the table at the right.

FITTING SIZE	POINT NUMBER AND OUTSIDE DIAMETER									
	PT	OD	PT	OD	PT	OD	PT	OD	PT	OD

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APPENDIX B—SAMPLE SEMIANNUAL STACK INSPECTION RECORD

The condition of a number of stacks can be tabulated on a form such as the sample contained in this appendix.

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SAMPLE SEMIANNUAL STACK INSPECTION REPORT

PLANT _____
CHECKED BY _____
REPORT BY _____
APPROVED BY _____
DATE _____

COMPLETE DESCRIPTION OF ALL DEFECTS AND REPAIRS SINCE LAST INSPECTION: *Inspected for condition and found OK except as noted below.*

STACK NO.	LOCATION AND DESCRIPTION	FOUNDATION	SHAFT	LINING	GUYS AND CONNECTION	LIGHTNING RODS, POINTS, CONDUCTORS, AND GROUNDS	LADDERS	VERTICAL ALIGNMENT	REMARKS AND RECOMMENDATIONS
30	Boiler No. 8, BH No. 2, 6 × 50 ft. SS	Blower housing	OK	None	None	None	None	OK	General condition good
31	Boiler No. 2, BH No. 2, 6 × 50 ft. SS	Blower housing	OK	None	None	None	None	OK	General condition good
32	Boiler No. 3, BH No. 2, 6 × 50 ft. SS	Blower housing	OK	None	None	None	None	OK	General condition good
33	Boiler No. 9, BH No. 2, 6 × 50 ft. SS	Blower housing	OK	None	None	None	None	OK	General condition good
34	Boiler No. 10, BH No. 2, 6 × 50 ft. SS	Blower housing	OK	None	None	None	None	OK	General condition good
35	FCCU, 11 × 120 ft. RBS	Concrete to rock	OK	Firebrick OK	None	5 Points 2 Grounds OK	Outside ladder irons	OK	General condition good
36	FCCU, 4 1/2 × 108 ft. SS	Platform	OK	None	None	None	20 ft. long	2 1/2 in. south, 10 in. west	General condition good
37	FCCU gas flare stack, 1 ft., 8 in. OD × 250 ft. high × 9/32 in.	Concrete to rock	OK	None	OK	None	None	3 ft. east, 20 in. south	Recently reconditioned–condition good: frozen concrete crumbling south side on pedestal
38	Badger pipe stills	Concrete	OK	OK	None	OK	Outside irons	OK	General condition good

Note: Stack numbers do not appear on any stack. Abbreviations are as follows: BH=blower housing; BS=brick stack; CS=concrete stack; SS=steel stack; RBS=radial brick stack; RTS=radial tile stack; FCCU=fluid catalytic cracking stack.

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Order No. 822-57300

1-1300—10/91—7.5C (9C)

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